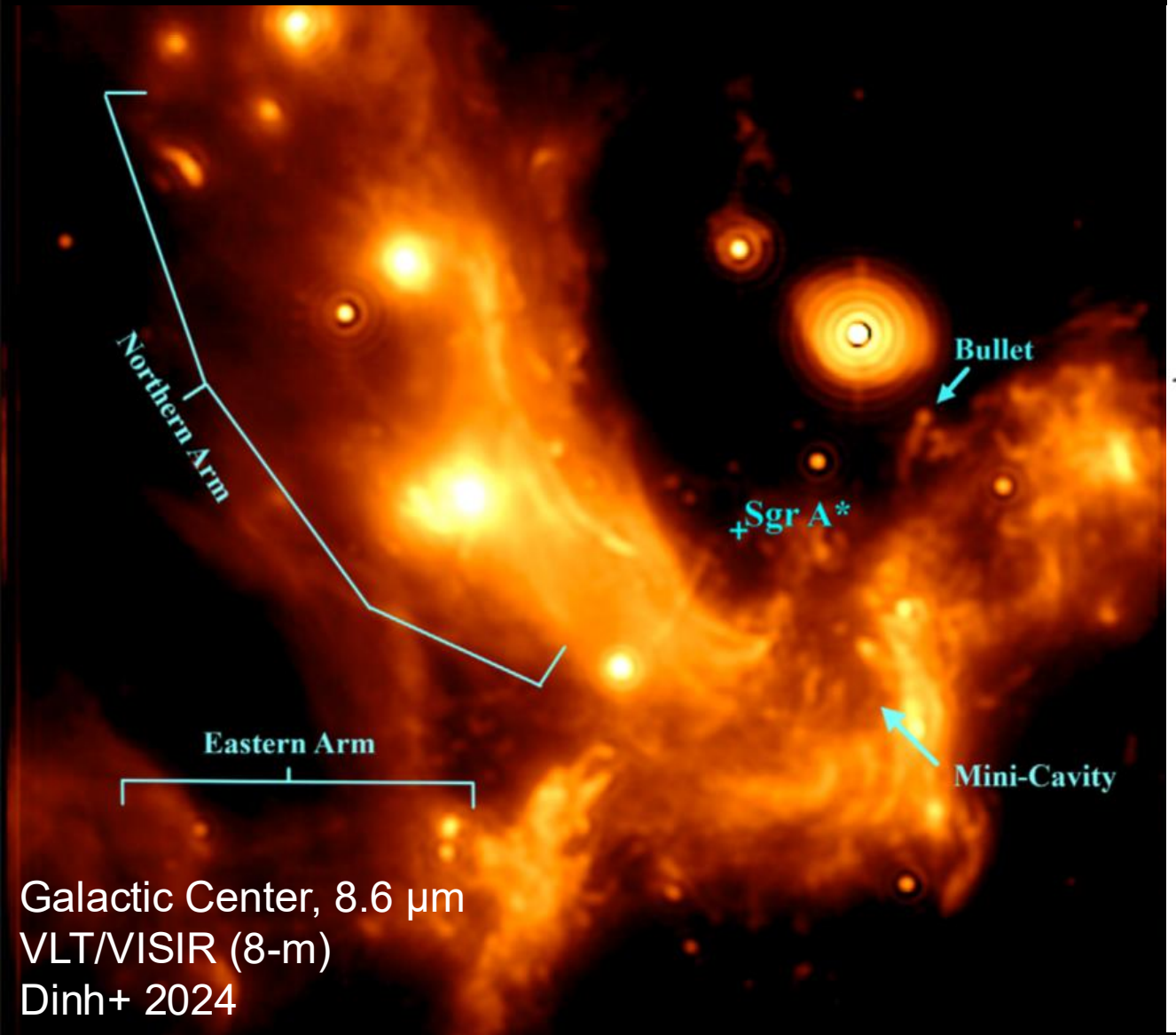
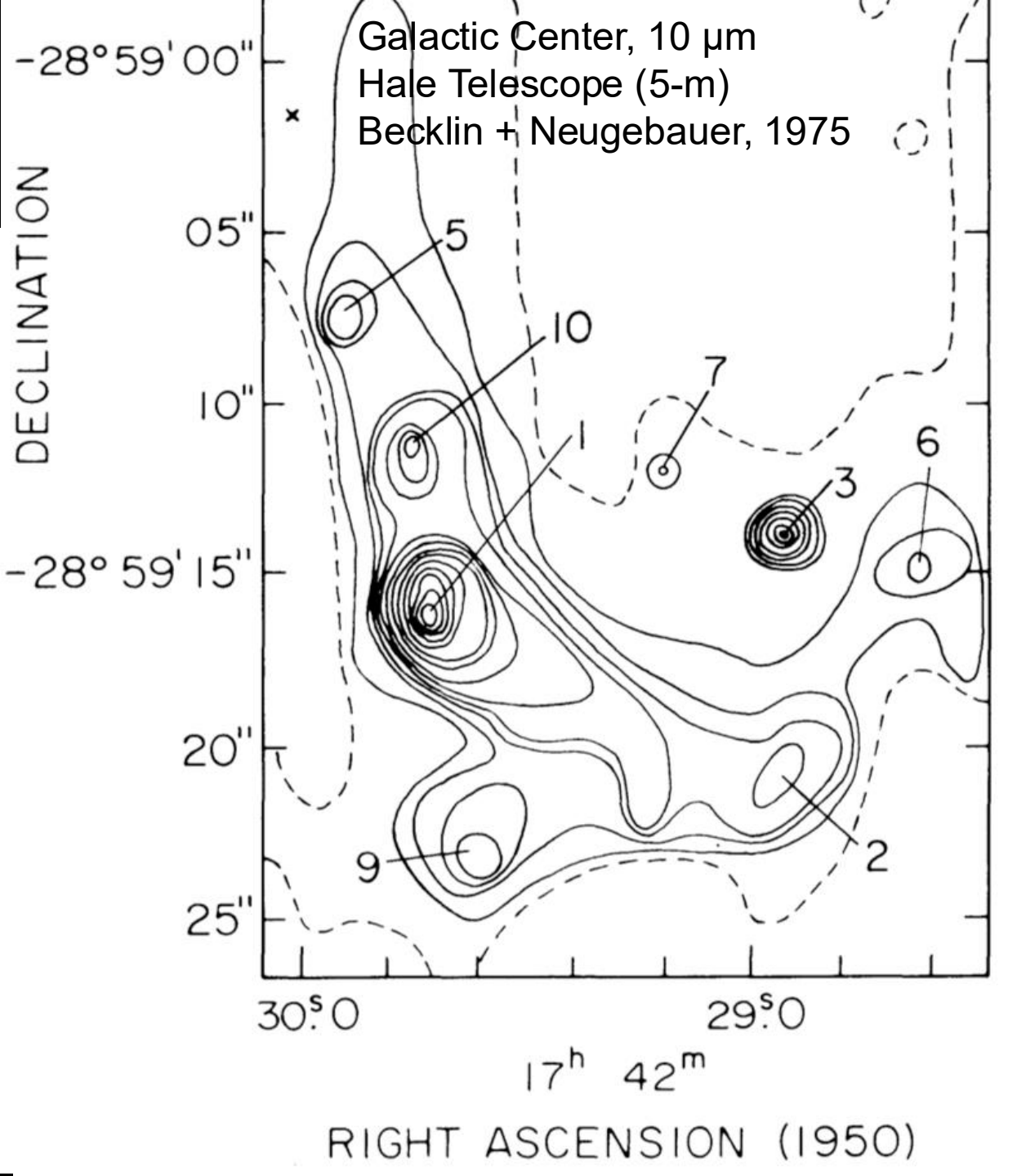
ASTR 302: Infrared Astronomy



Galactic Center, 8.6 μm
VLT/VISIR (8-m)
Dinh+ 2024

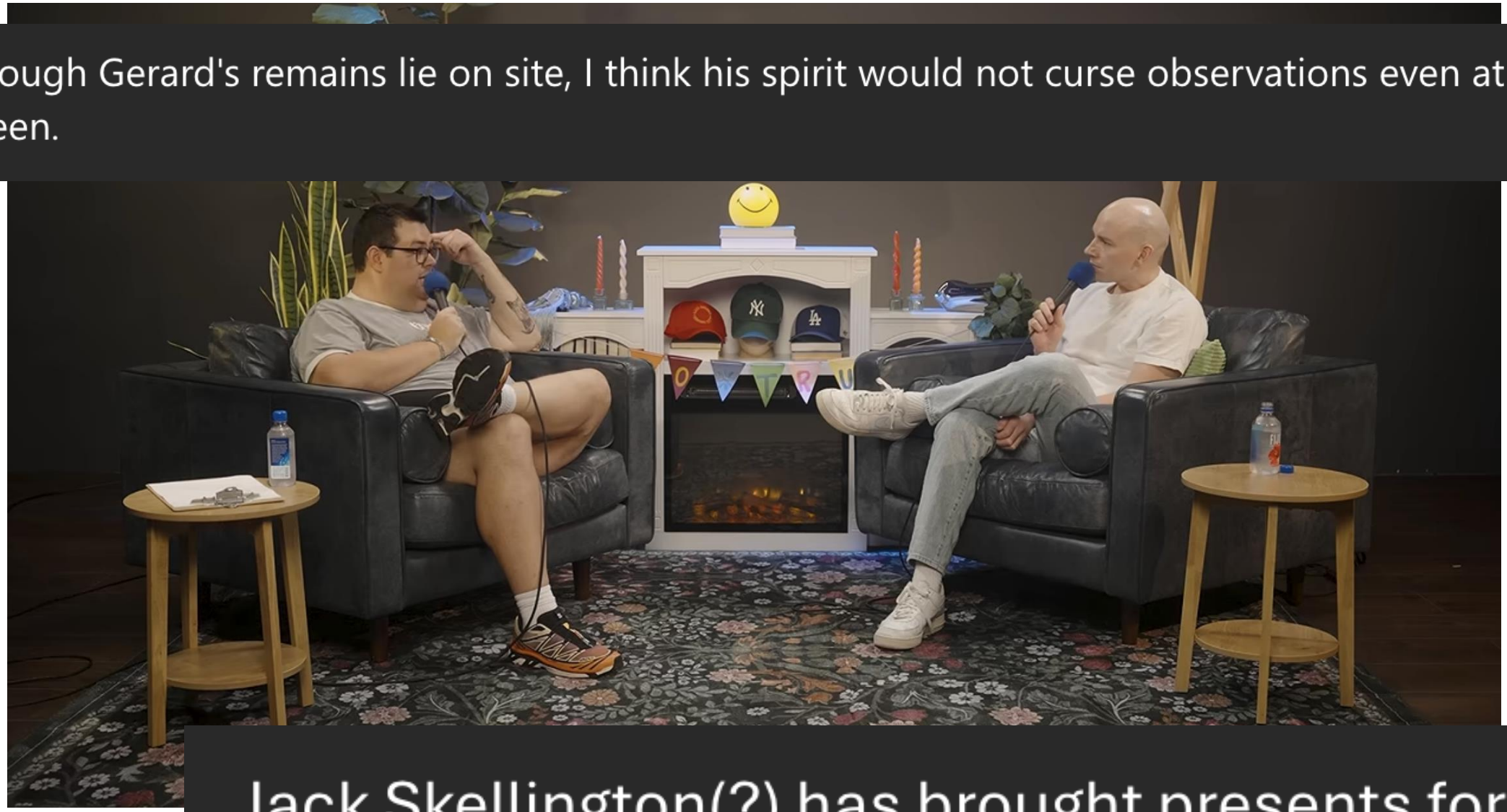


Agenda – October 26th, 2026

- Weekend Recap
- Scientific Reading and Writing (Be in class Nov 4th *)
- Infrared Astronomy
  - Detectors
  - Observational Strategies
- ~~Exoplanets and Brown Dwarfs~~

Weekend RECAP: ZACH GOT DATA!!!!

Even though Gerard's remains lie on site, I think his spirit would not curse observations even at Halloween.



Jack Skellington(?) has brought presents for all.

- **Mock Journal Club/Reading Day – November 4th , 2026**

BRING LAPTOP!!!!!!!!!!!!

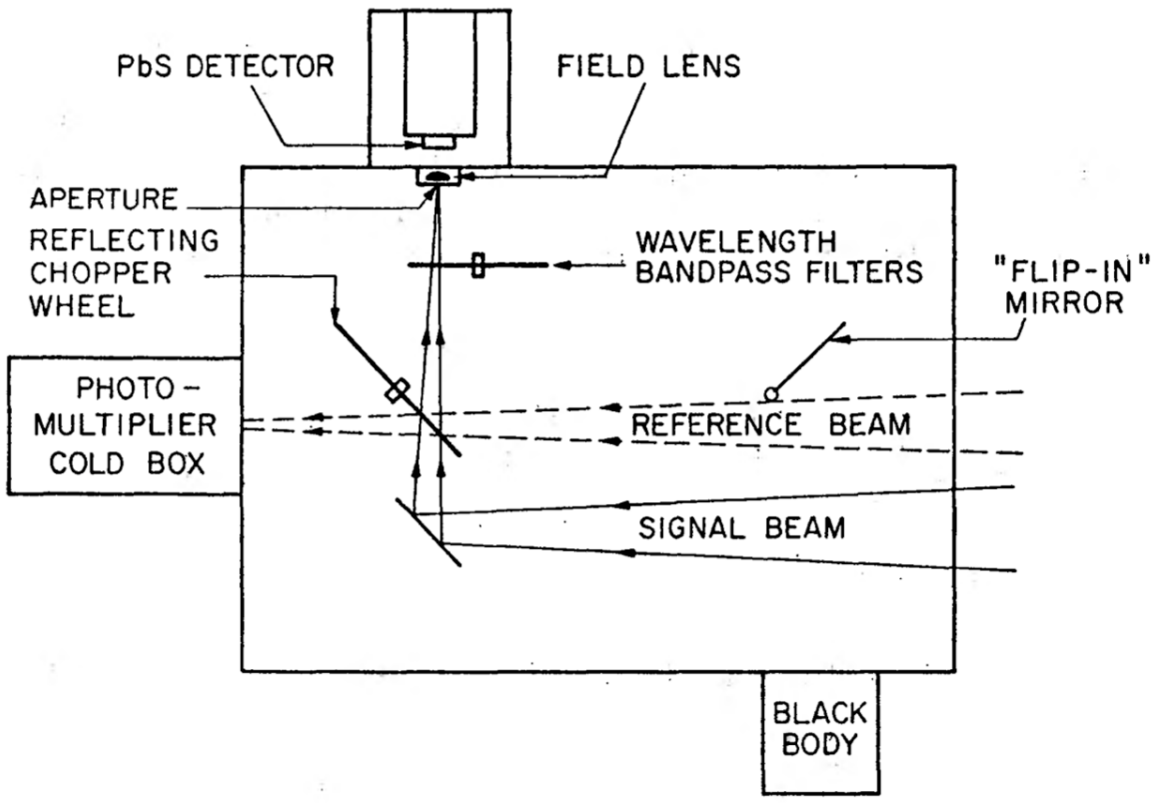
  - "The Discovery of a Very Cool, Very Nearby Brown Dwarf in the Galactic plane" – Lucas+ 2010 **(Imaging + Spectroscopy)**
  - "The Discovery of a Luminous Z=5.80 Quasar from the Sloan Digital Sky Survey" – Fan+ 2000 **(Imaging + Spectroscopy)**
  - "High Proper-Motion Stars in the Vicinity of Sagittarius A*: Evidence for a Supermassive Black Hole at the Center of Our Galaxy" – Ghez+ 1998 **(Imaging)**

Agenda – October 26th, 2026

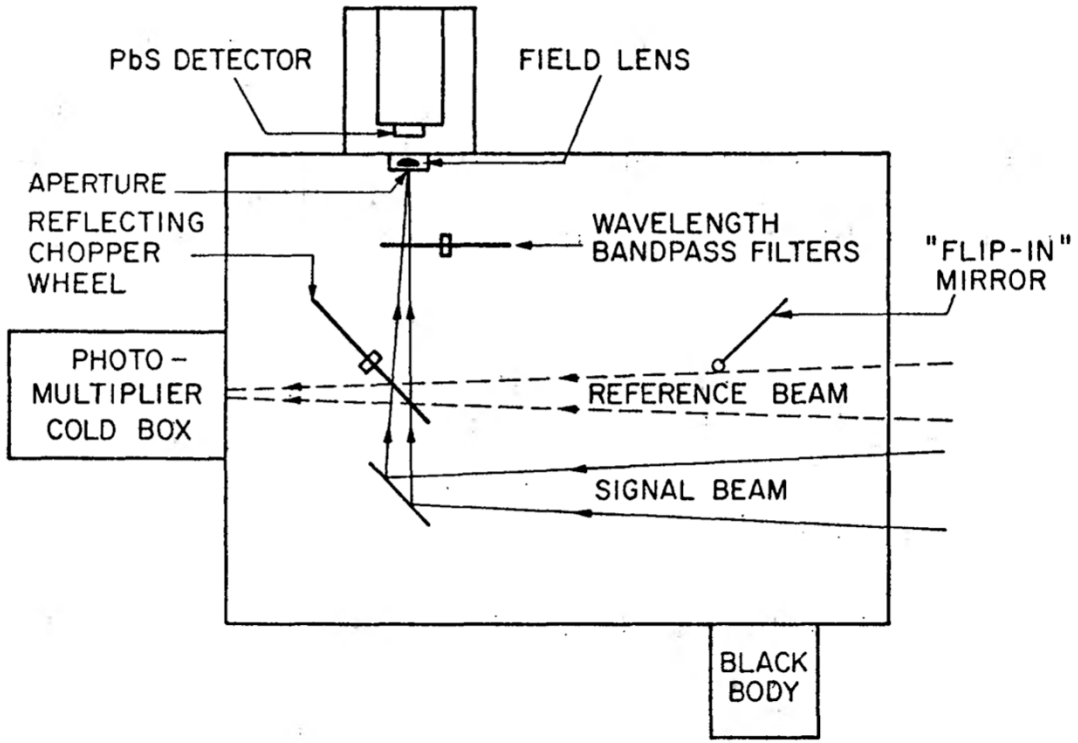
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Early Days of Infrared Astronomy

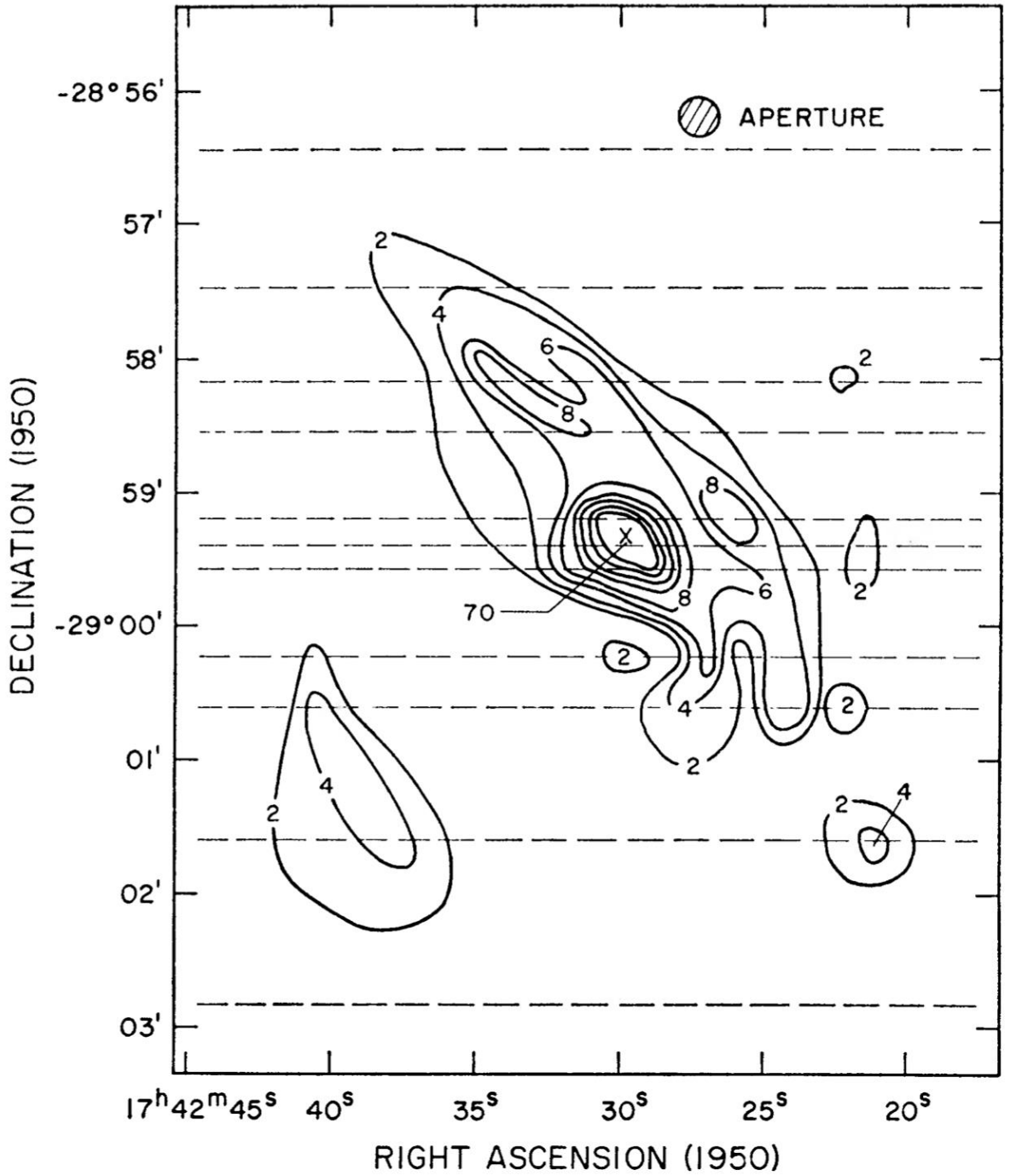
- Photographic plates were not commonly used in the infrared.
- You often had 1-pixel and you were grateful for it.
  - Thermocouples
  - Bolometers
  - PbS (Lead Sulfide) Detectors
- There are many people at Steward who saw infrared astronomy go from 1-pixel to 4k pixels.

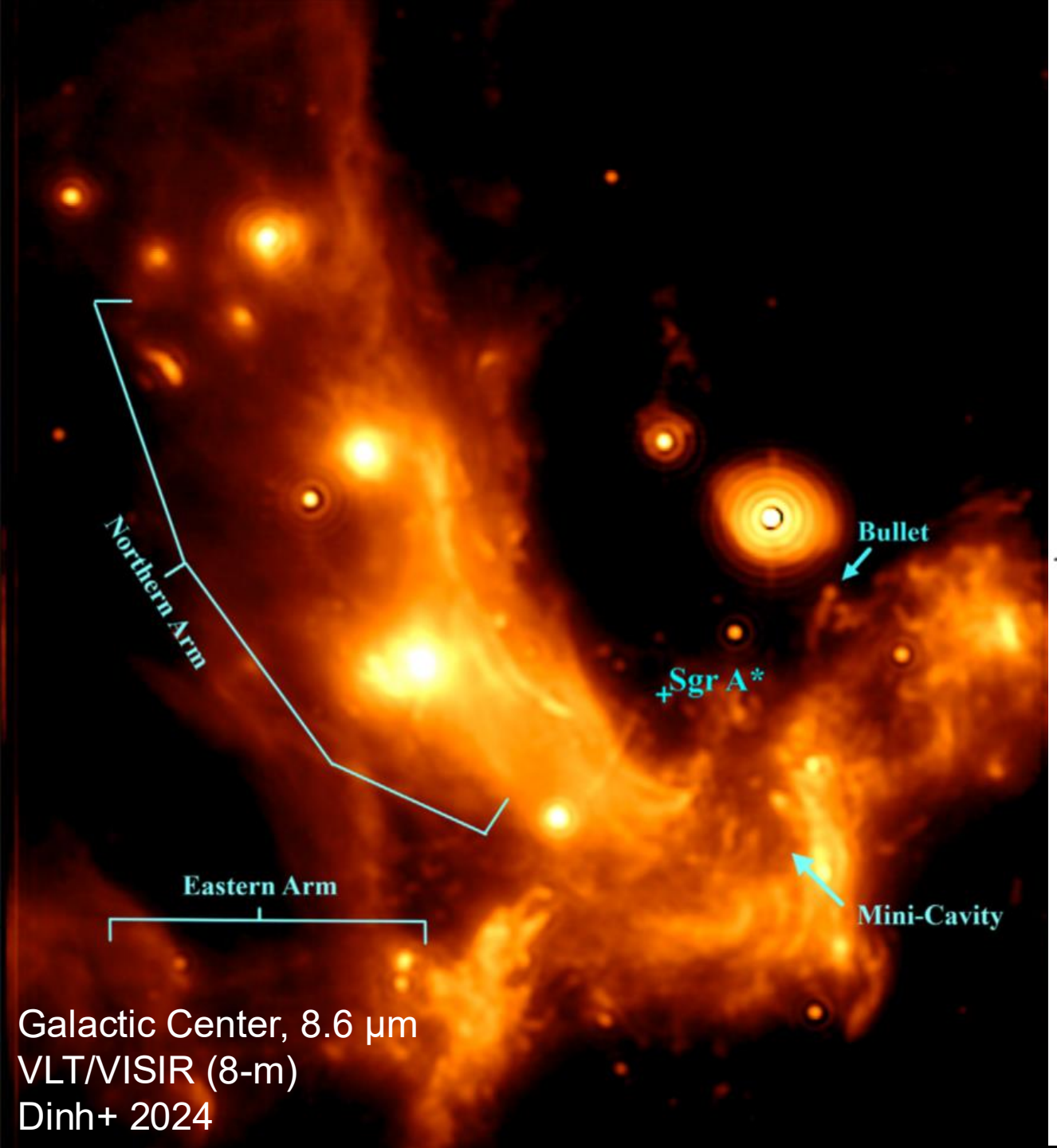


https://www.ipac.caltech.edu/Outreach/Edu/Timeline/timeline2.html

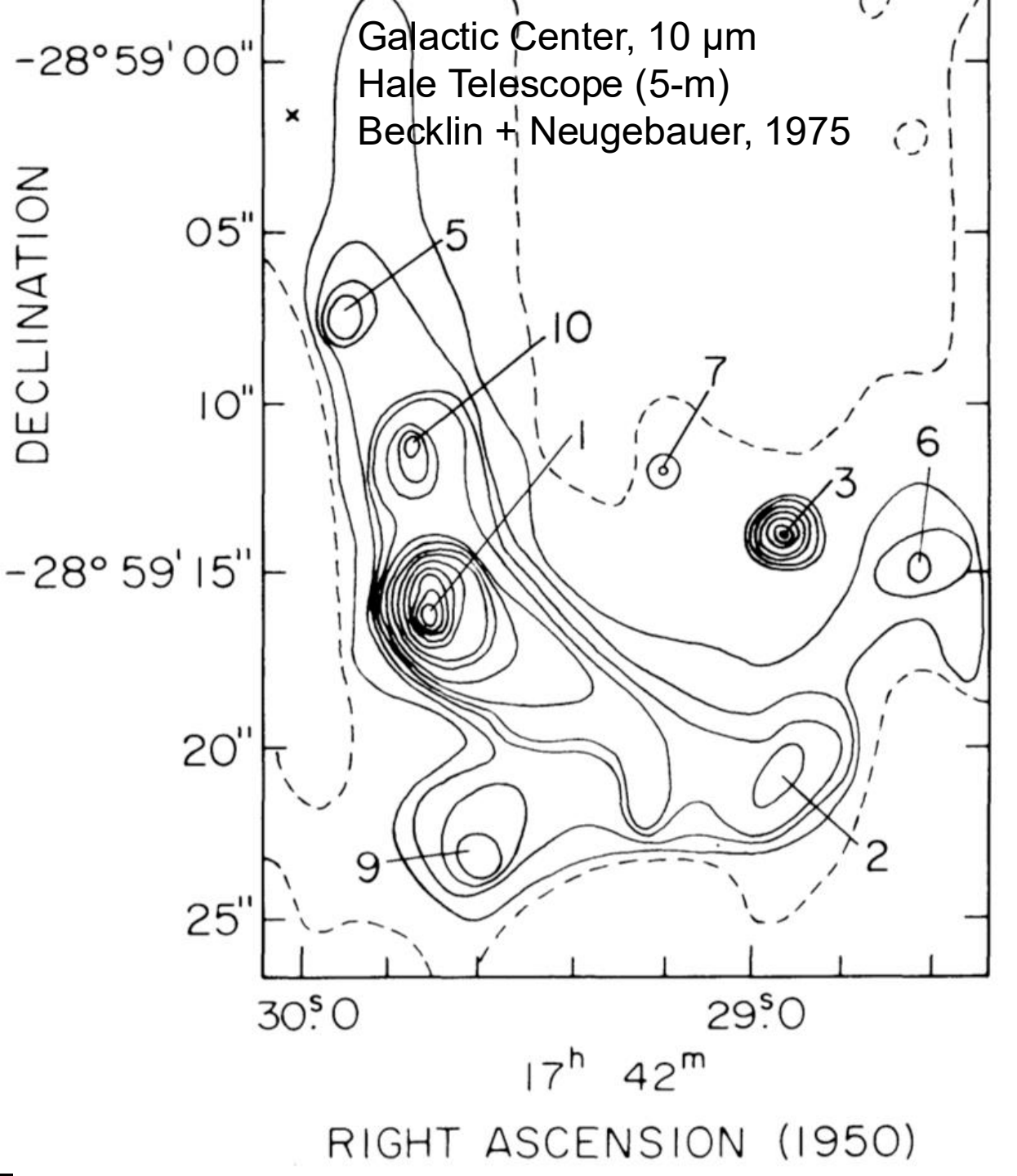


Becklin + Neugebauer, 1975

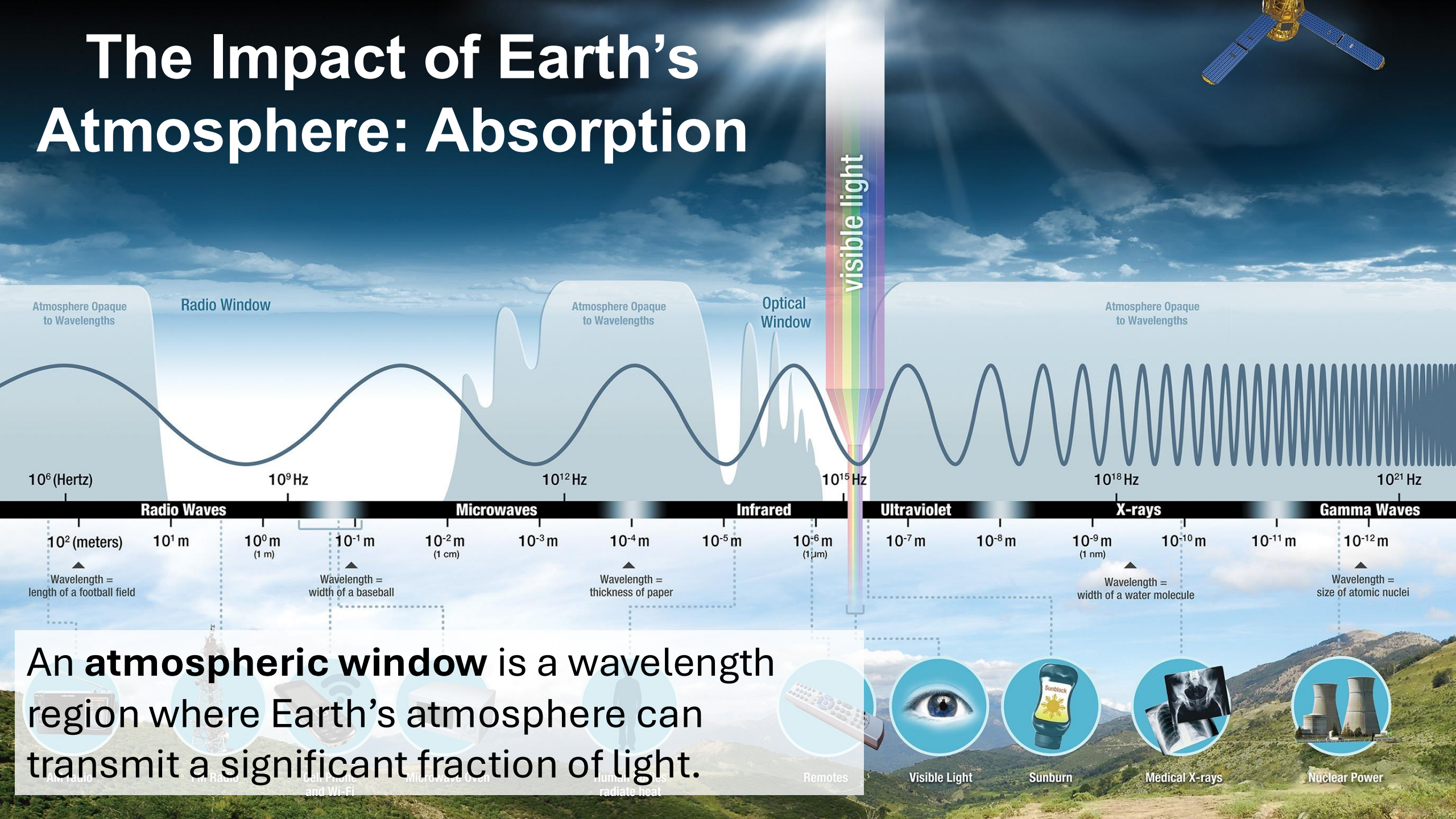




Galactic Center, 8.6 μm
VLT/VISIR (8-m)
Dinh+ 2024

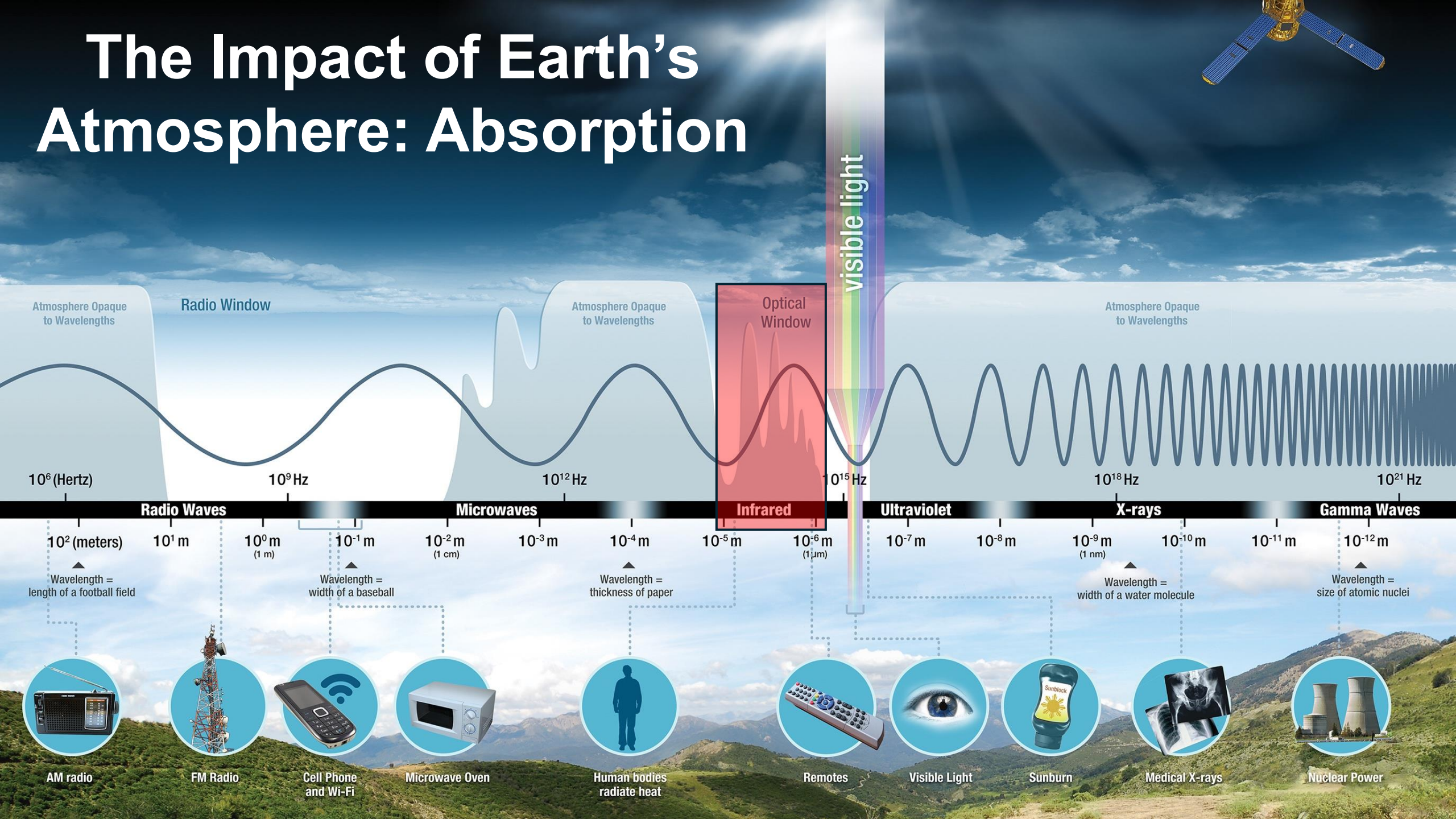


The Impact of Earth's Atmosphere: Absorption



An **atmospheric window** is a wavelength region where Earth's atmosphere can transmit a significant fraction of light.

The Impact of Earth's Atmosphere: Absorption



Infrared Wavelengths

Wavelength Range: 1 - ~1000μm

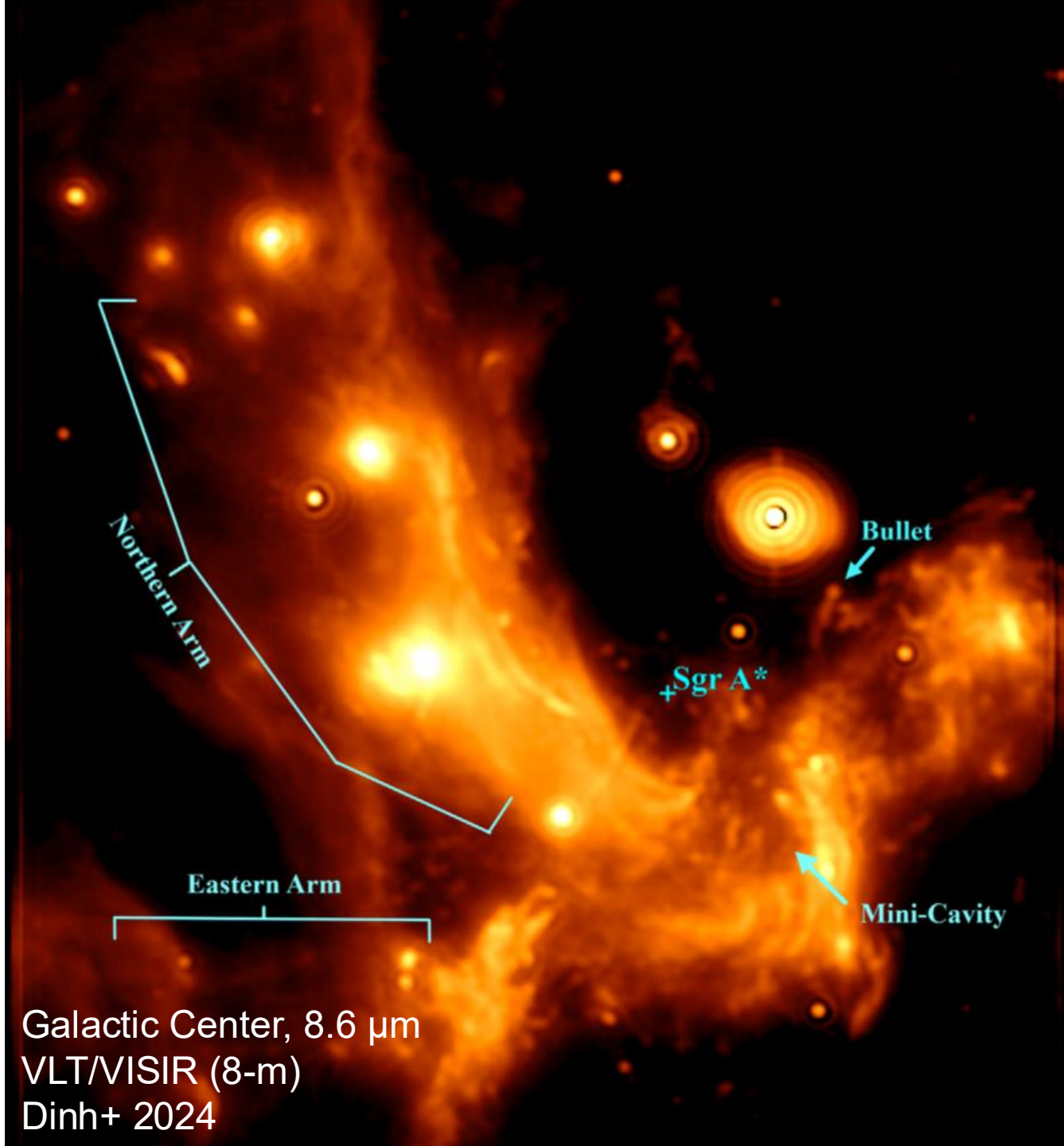
Near-Infrared: 1 – 2.5 μm

Mid-Infrared: 3 – 5 μm

Long-Wave Infrared: 8 – 14 μm

Far-Infrared: >15 μm

Many of these boundaries are motivated by atmospheric windows within Earth's atmosphere.



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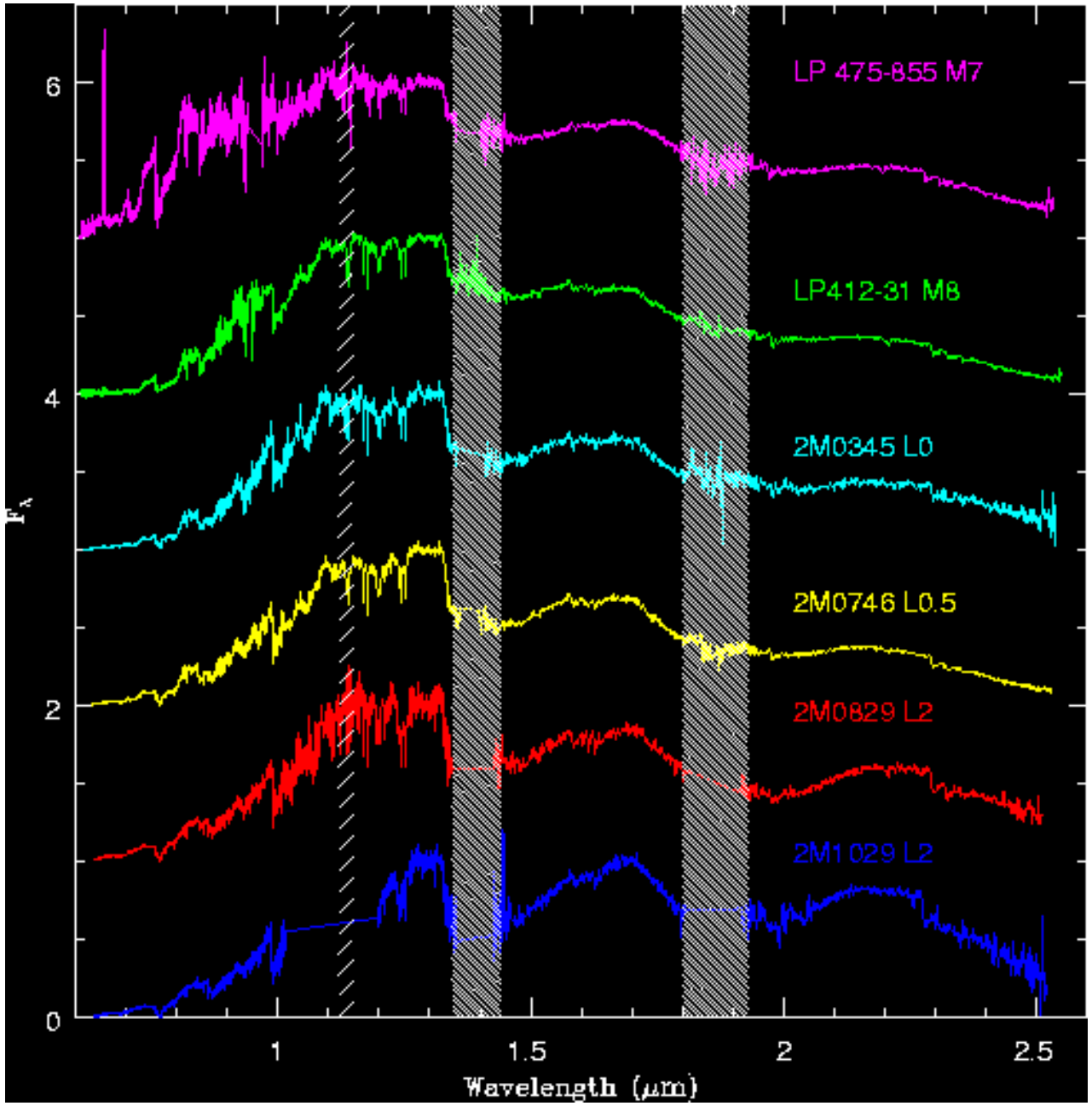
Near-Infrared: 1 – 2.5 μm

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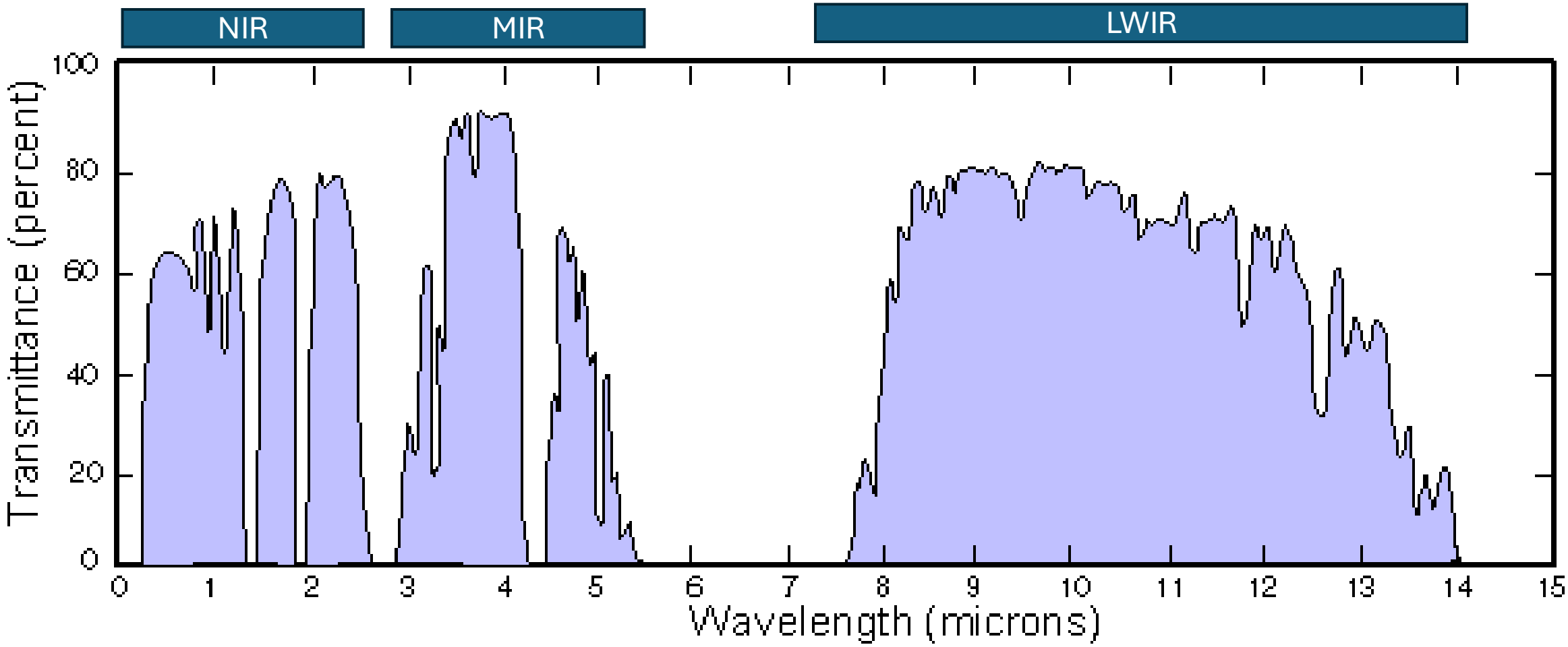
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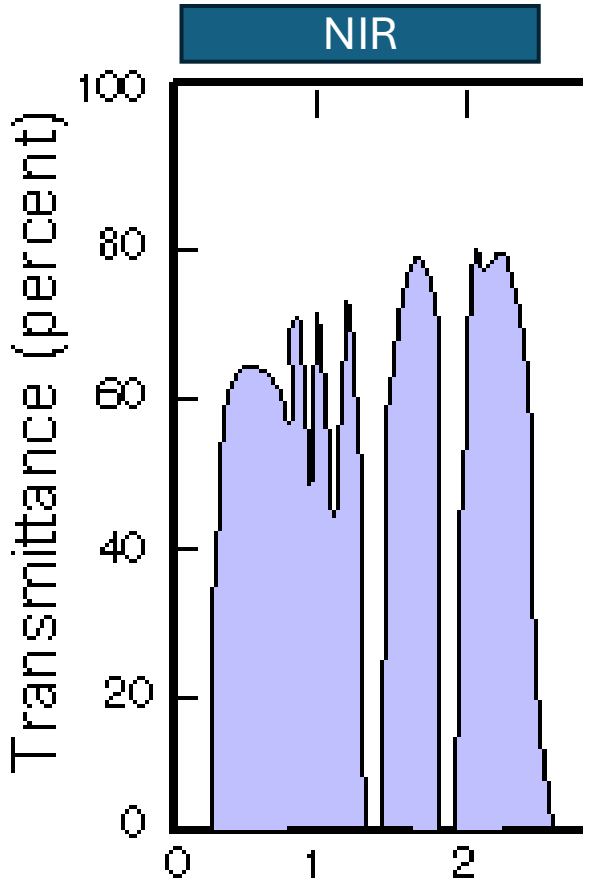


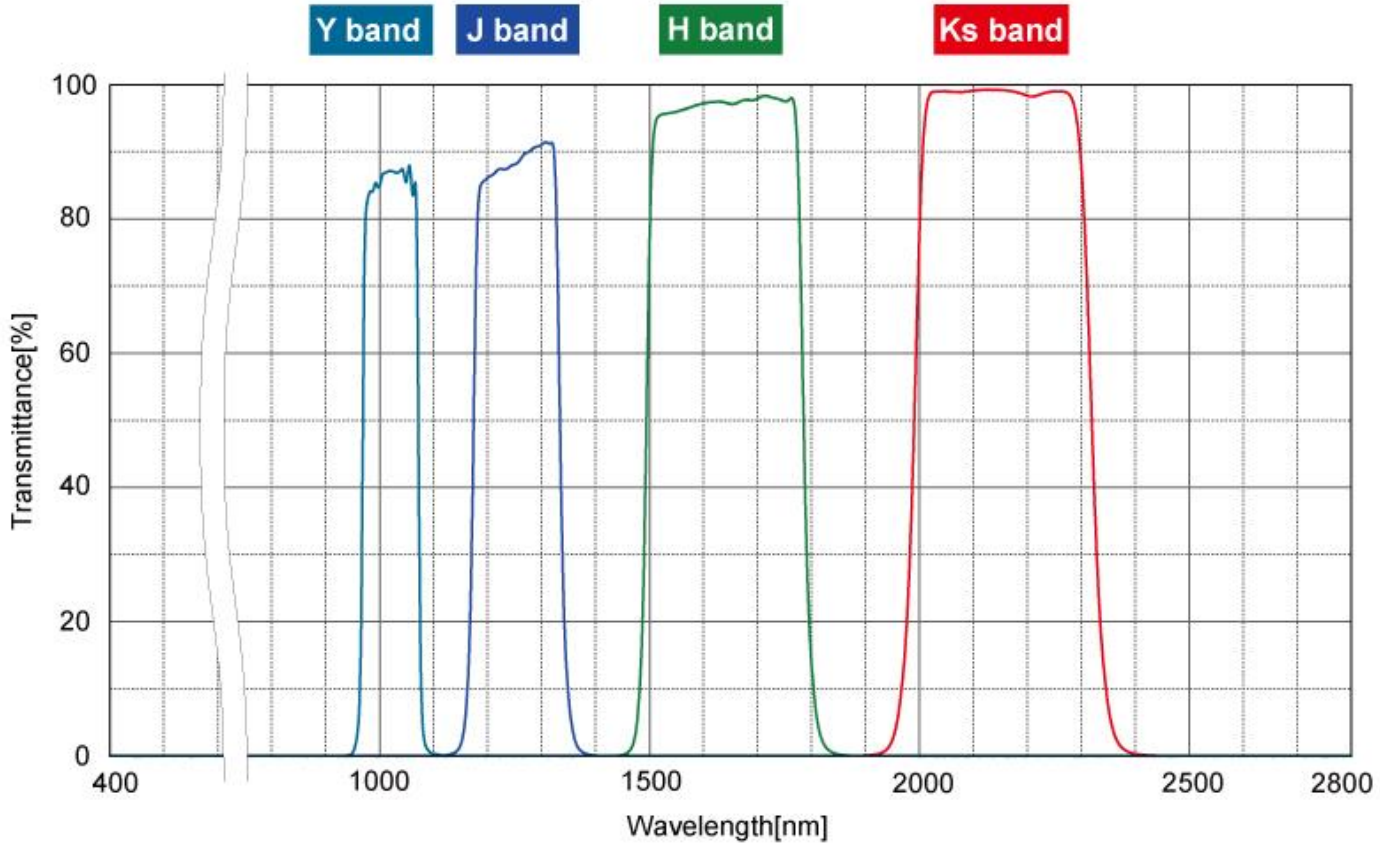
Infrared Atmospheric Windows



Many of these IR region boundaries are motivated by atmospheric windows within Earth's atmosphere. IR filters are often optimized to these regions.

Infrared Atmospheric Windows

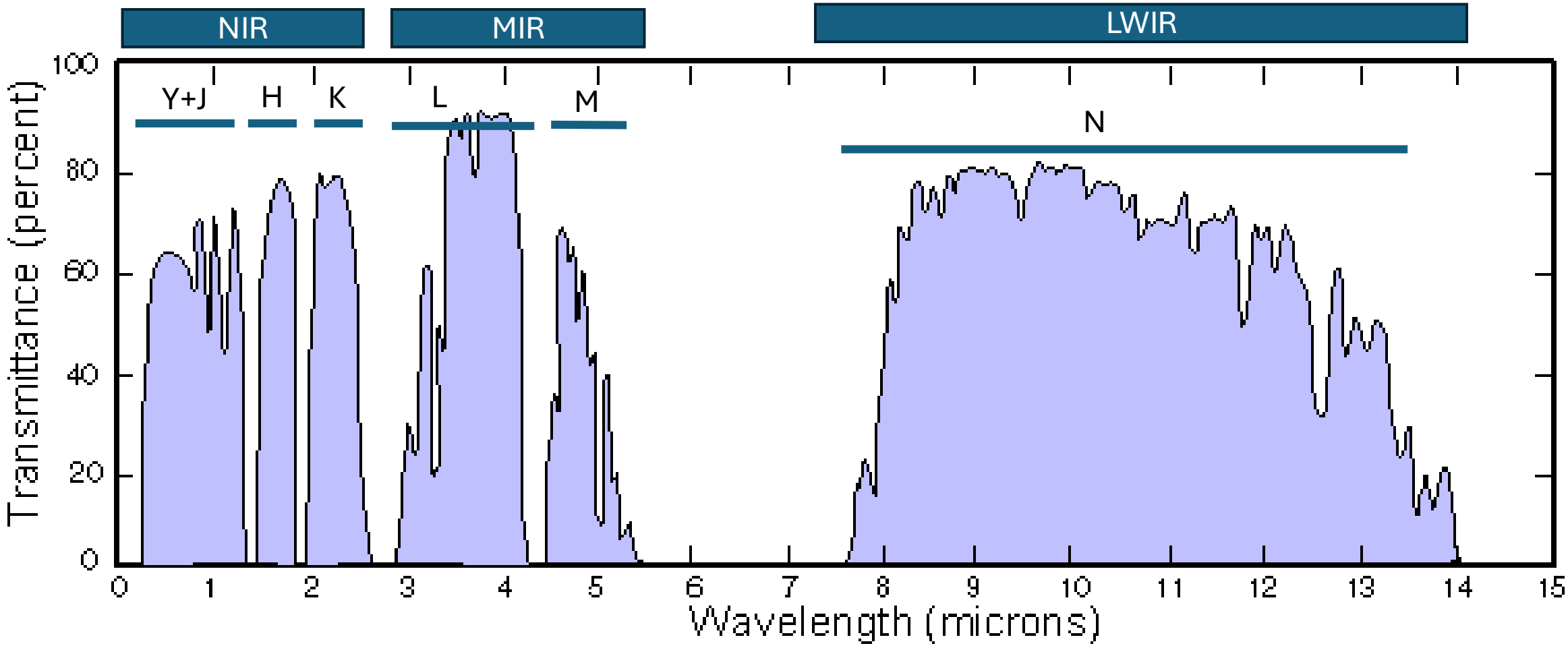




Asahi Spectrum (company)

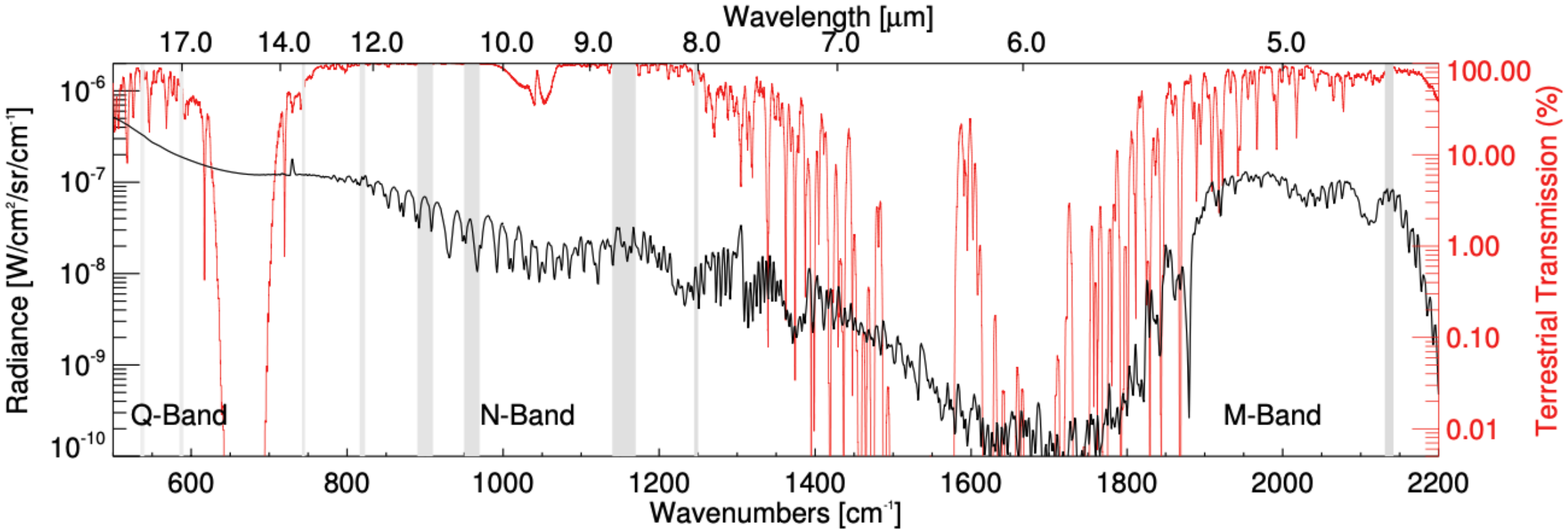
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Infrared Atmospheric Windows



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Infrared Atmospheric Windows

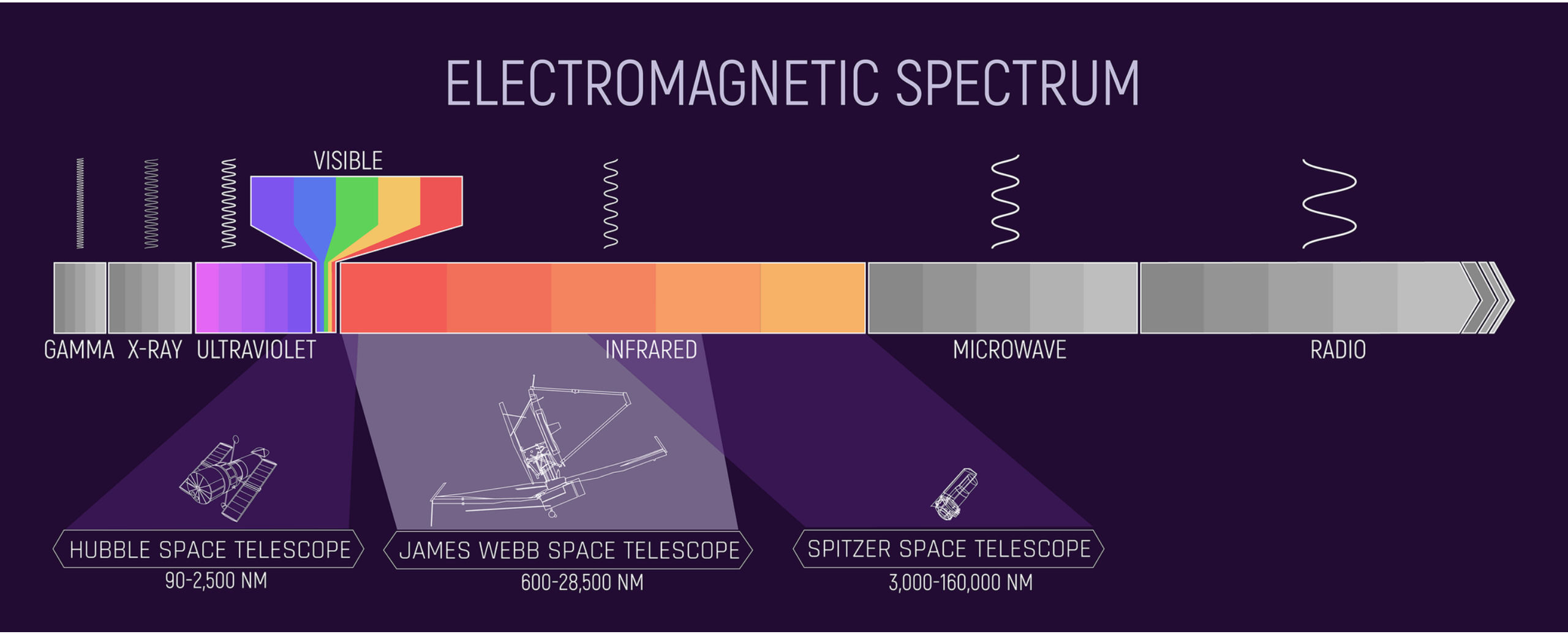


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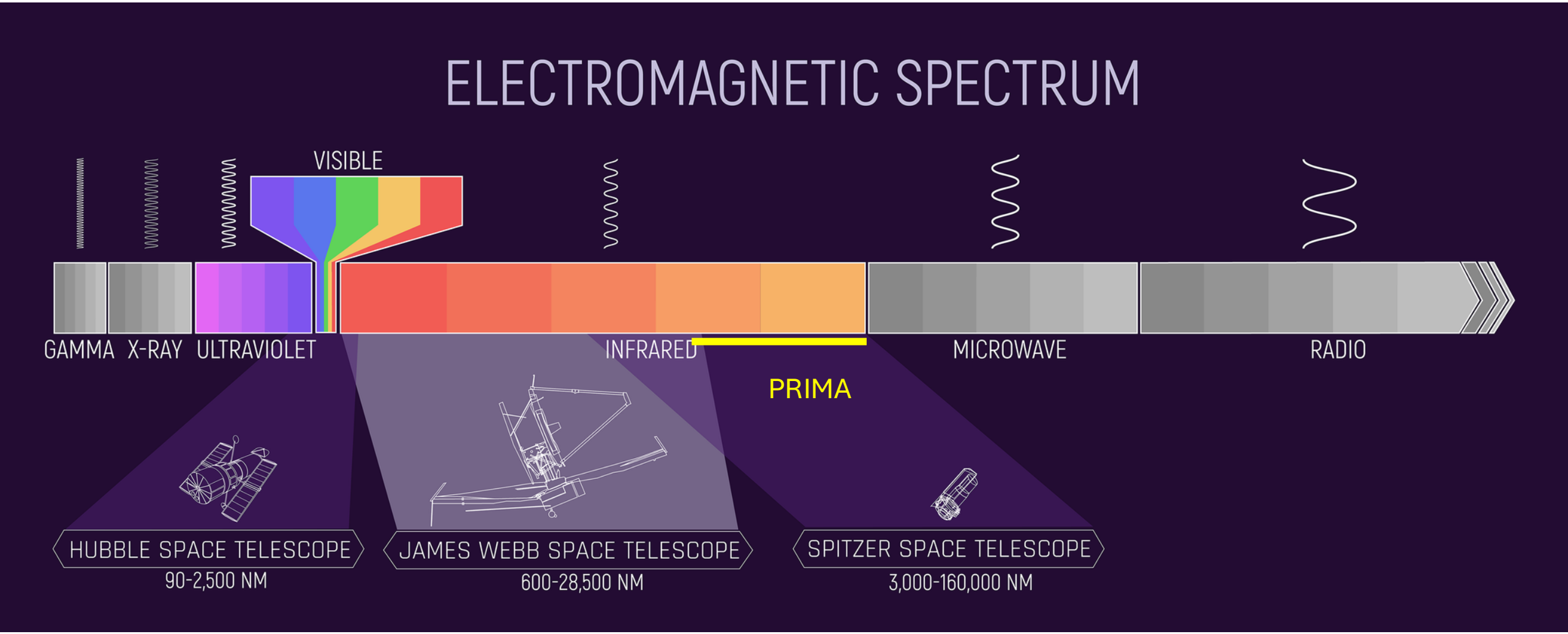
Q-band (17-25 μm)

Fletcher+ 2016 (Jupiter Study)

Infrared Wavelengths: Space-Based Observatories



Infrared Wavelengths: Space-Based Observatories



Detecting and Recording Light: CCD pixels

Single CCD Pixel

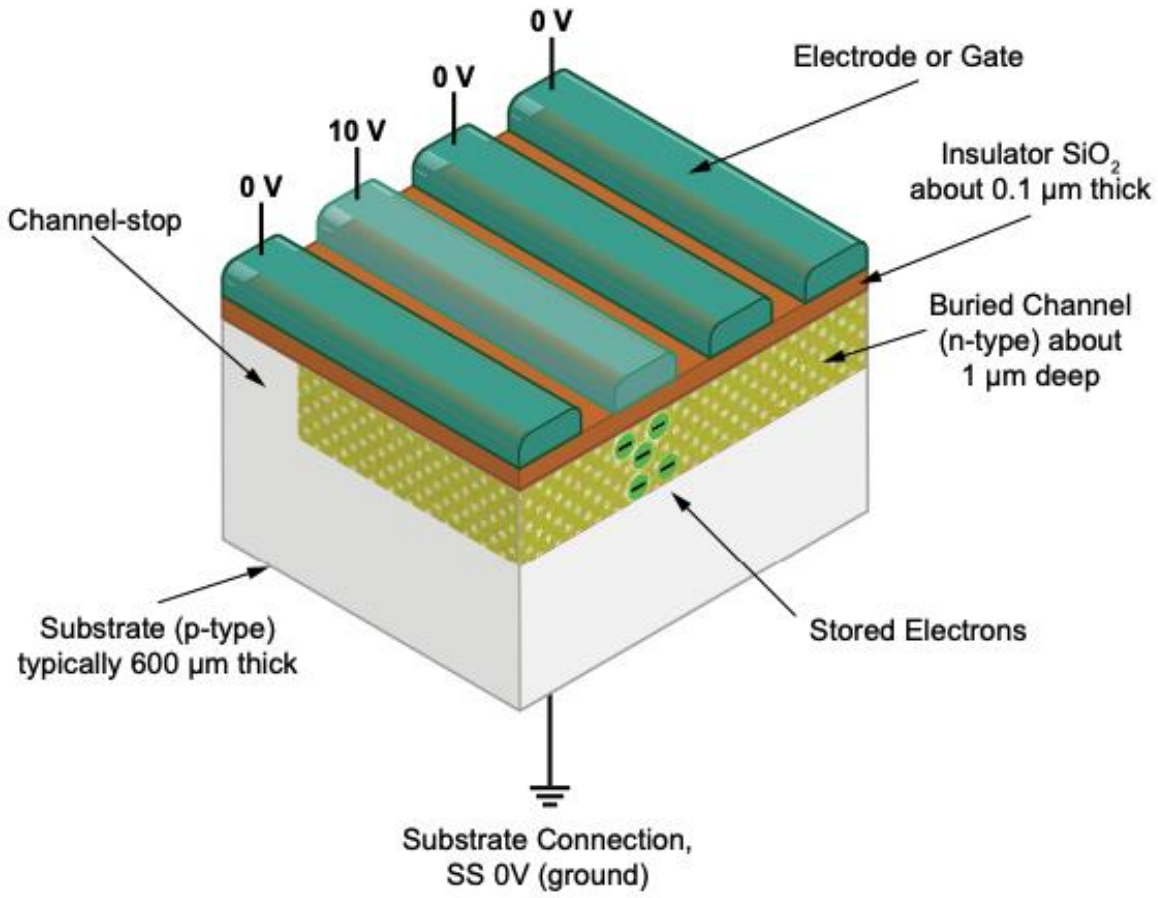
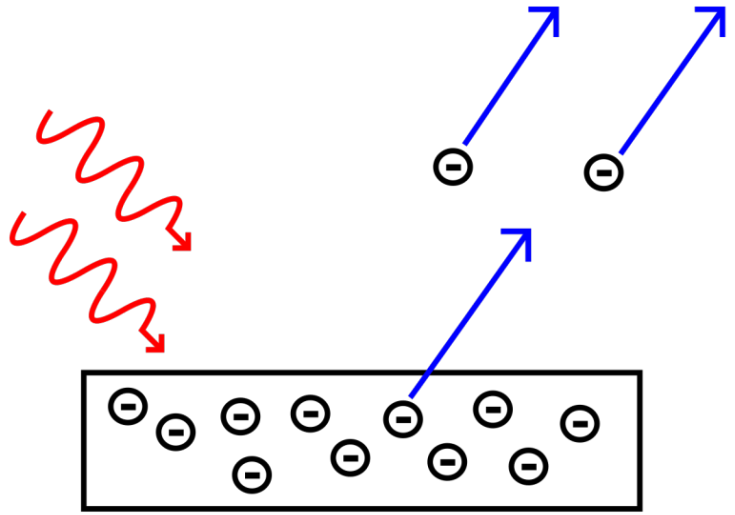


Image from Teledyne Imaging

Photoelectric Effect



- Because of the photoelectric effect, photons hitting the substrate releases electrons.

Detecting and Recording Light: Infrared

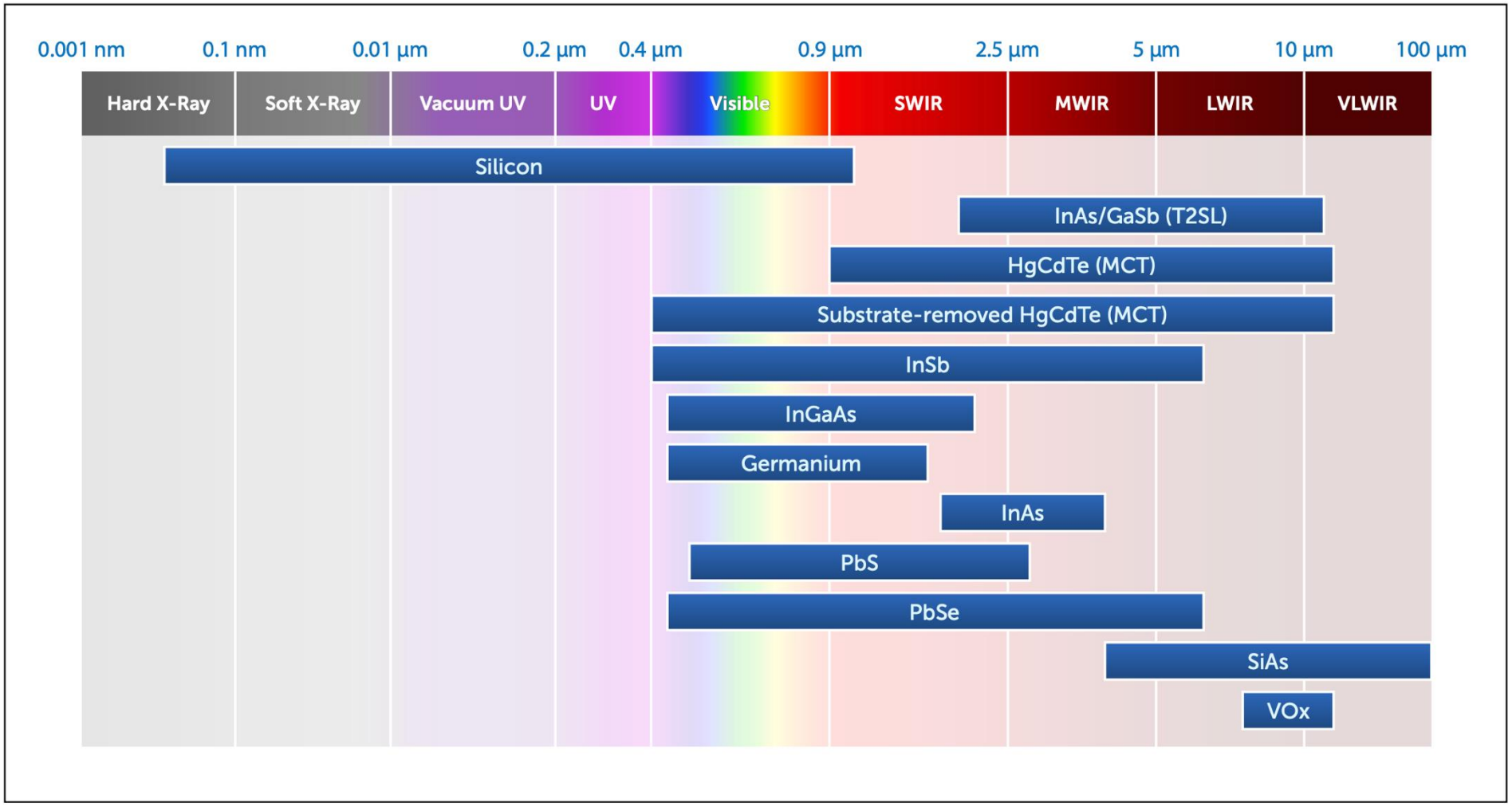
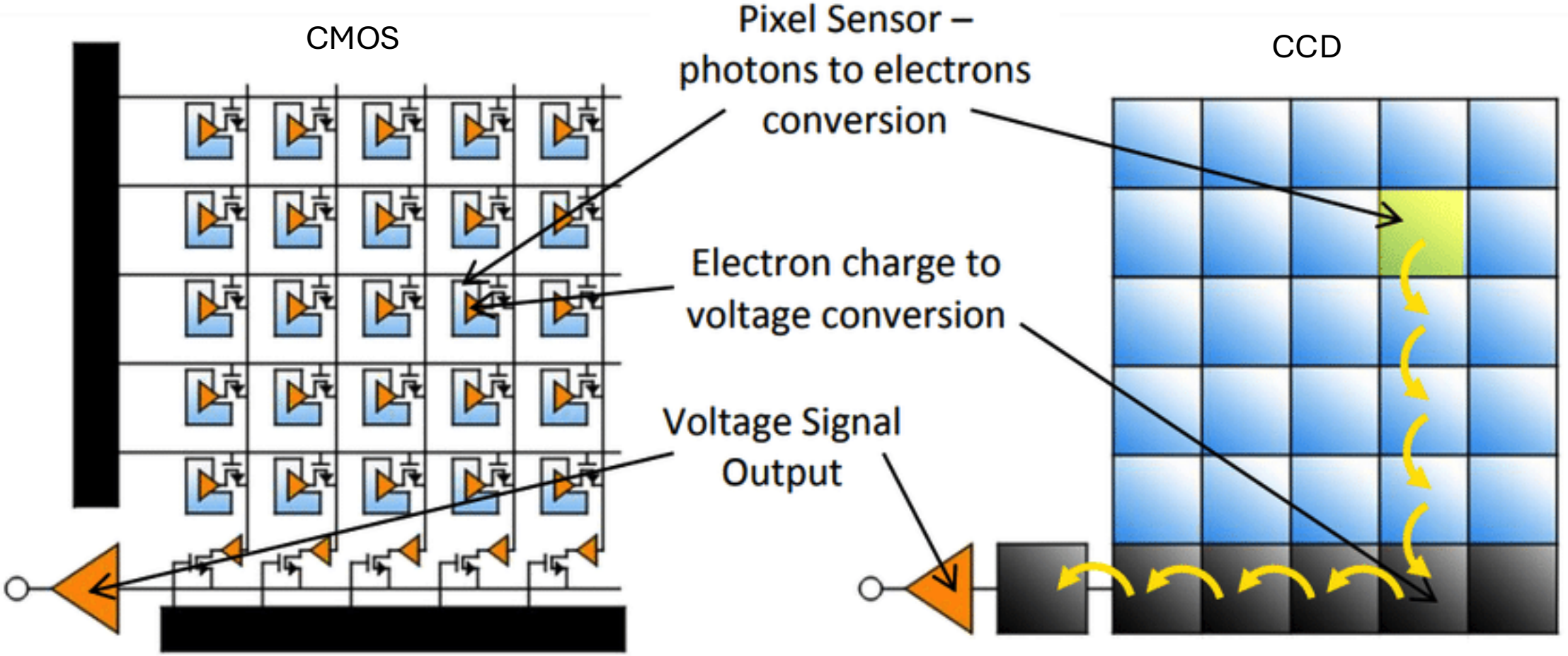


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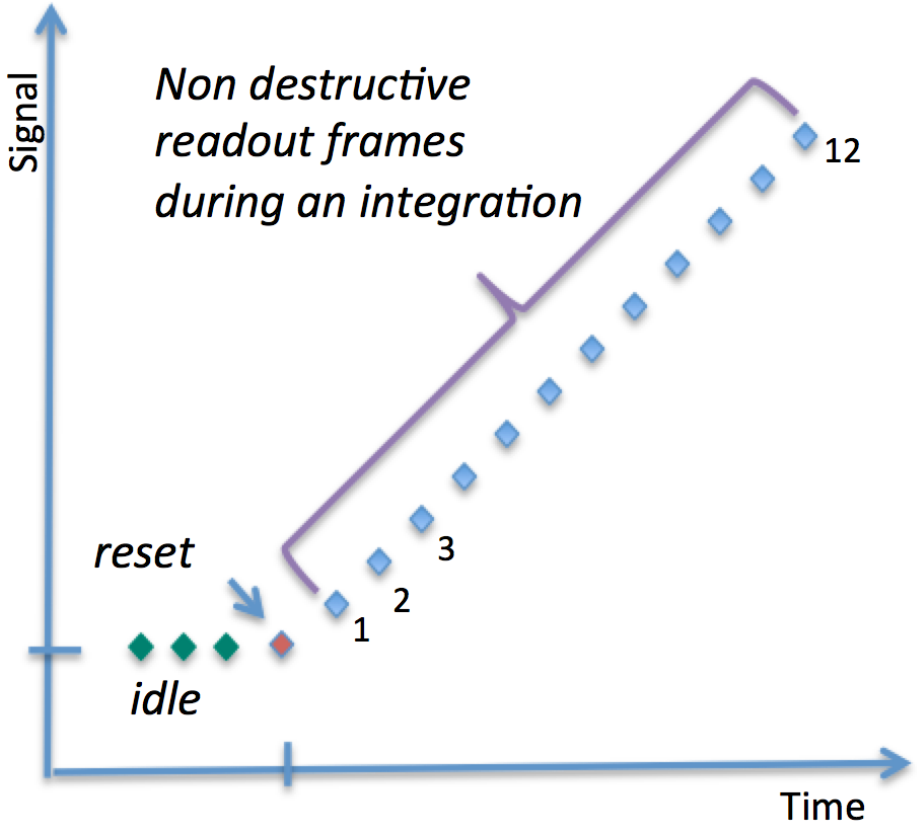
The band gap energy of Silicon (1.11 eV) is too high for lower energy light to release electrons. We often use other materials.

CMOS Read Out vs CCD Read Out



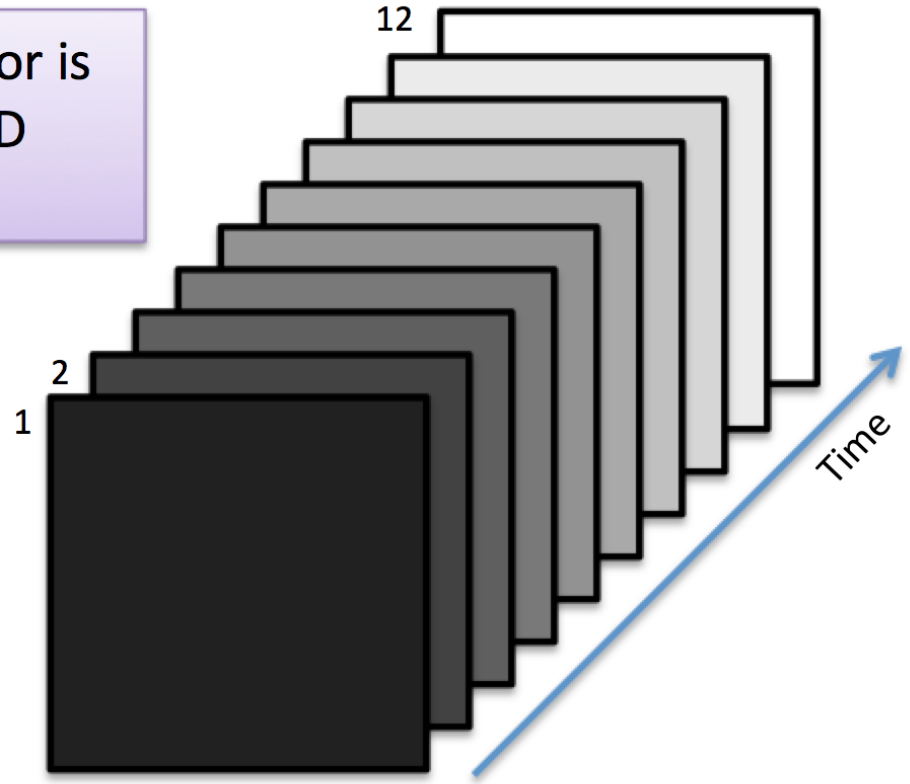
CMOS (Complementary Metal-Oxide-Semiconductor) detectors are used in your phones and other industrial applications. Some astronomical detectors use CMOS or CMOS-like readouts. Supposedly ~1000 times faster (Teledyne Vision Systems)

Non-Destructive Reading on Detectors



Integration timeline

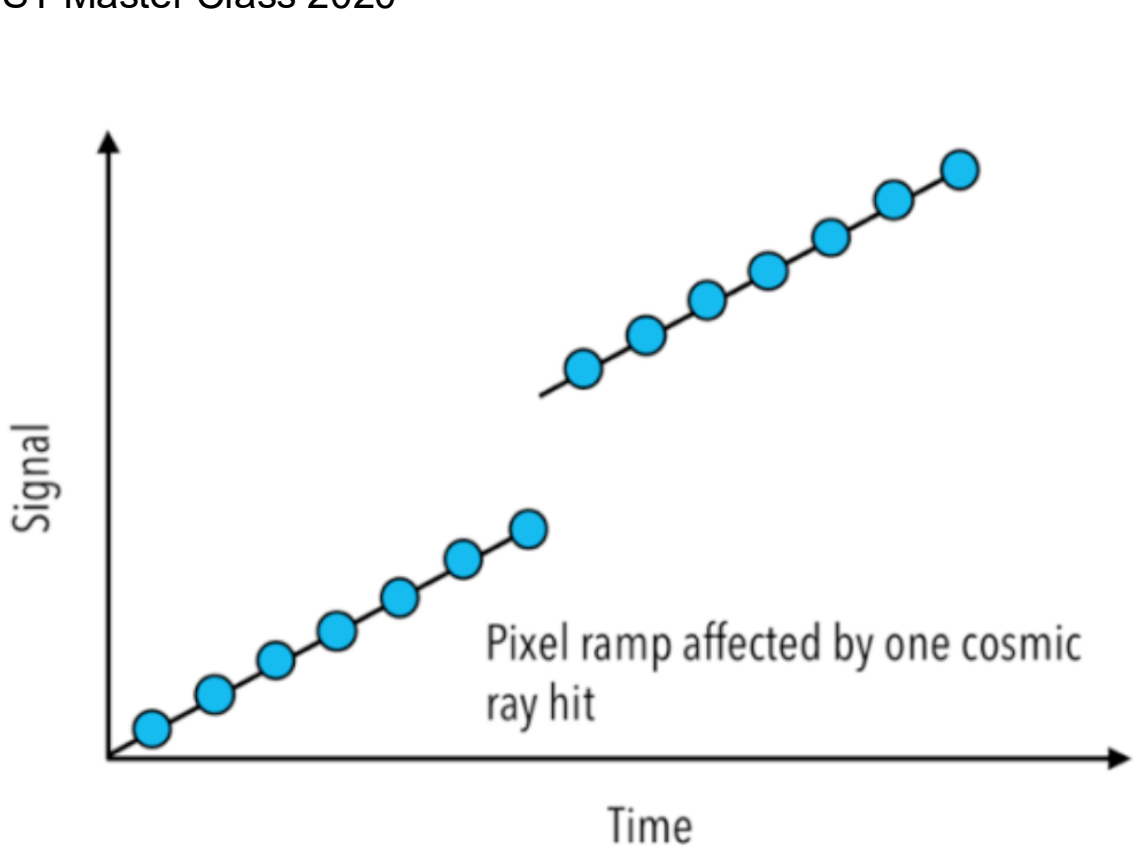
Each read of the detector is saved as a slice in the 3D data cube

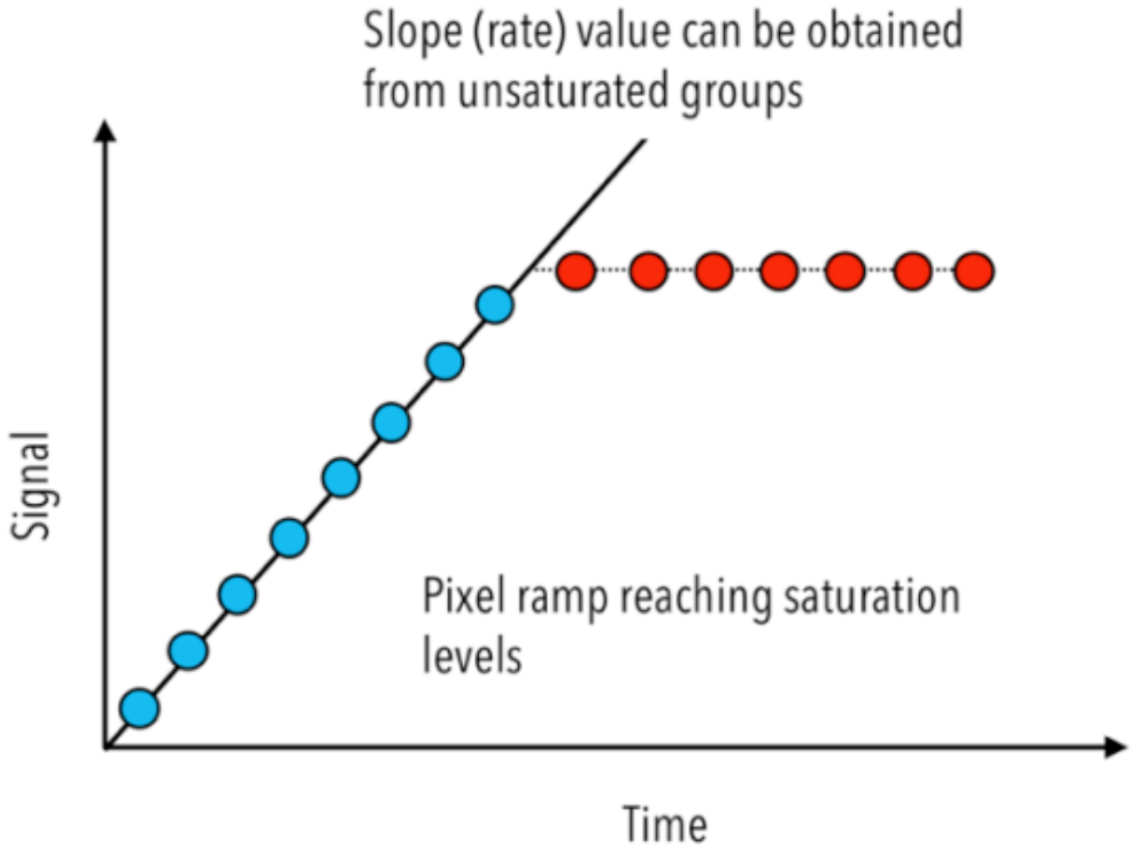


Datacube Structure

Non-Destructive Reading on Detectors

JWST Master Class 2020

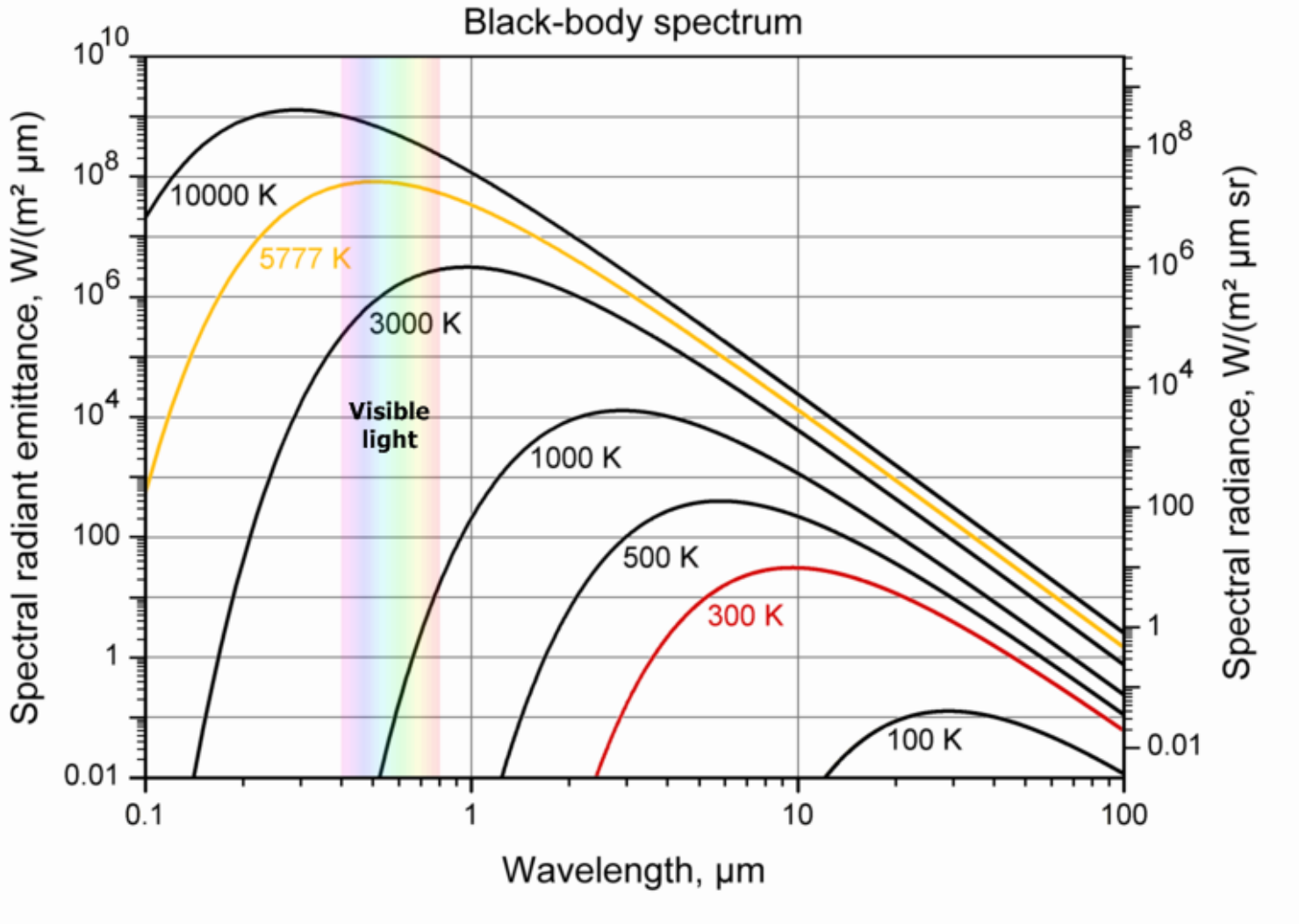




Reading up the ramp allows you to see when cosmic rays hit. You can also retain data before the pixel becomes saturated. You can preserve some response. information.

Infrared Astronomy Requirements

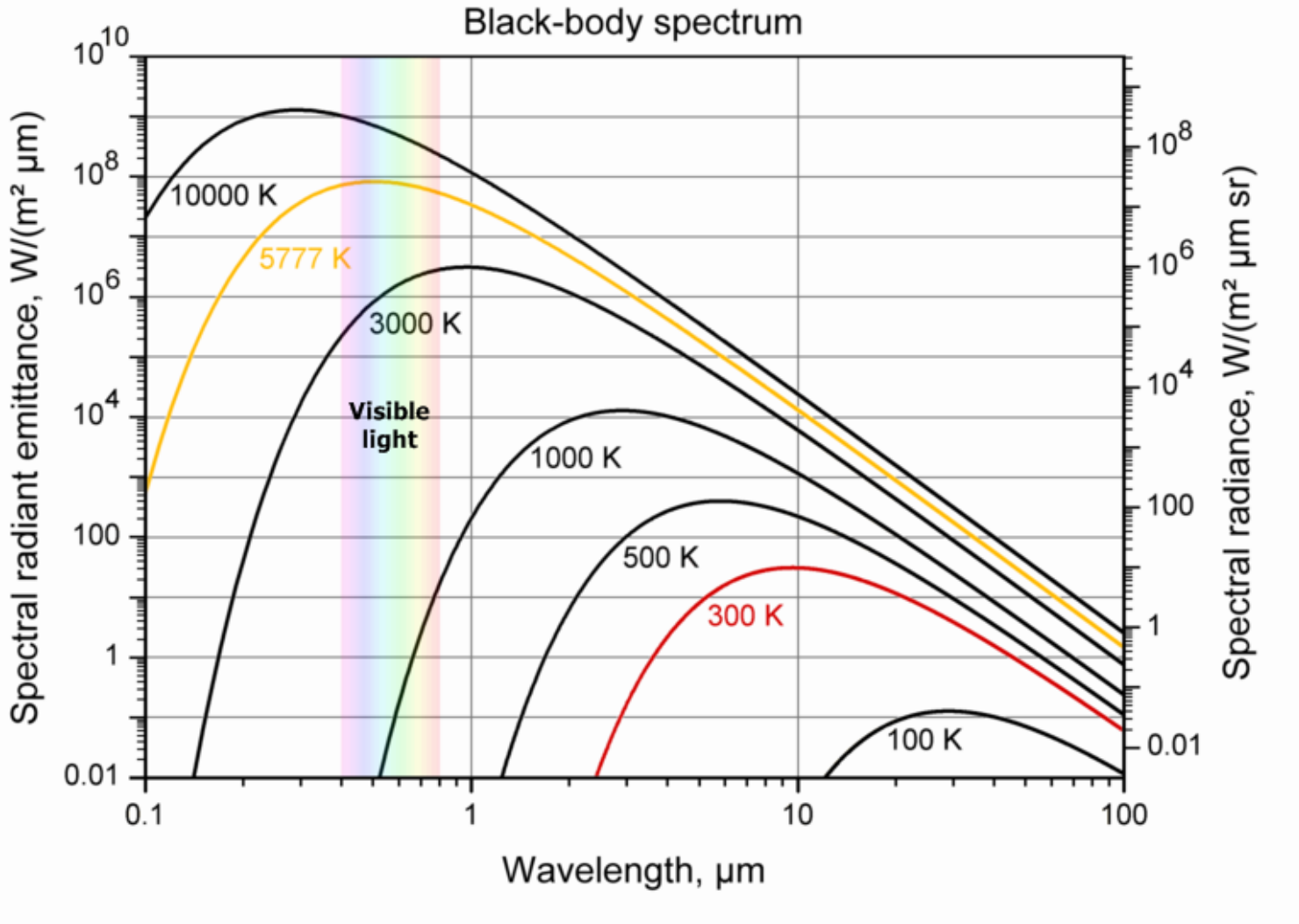
The photosensitive materials for IR detectors have such a low band gap energy they must be cooled to limit dark current.



Infrared Astronomy Challenges

The black body response of any surface at room temperature can significantly impact mid to long-wavelength infrared observations.

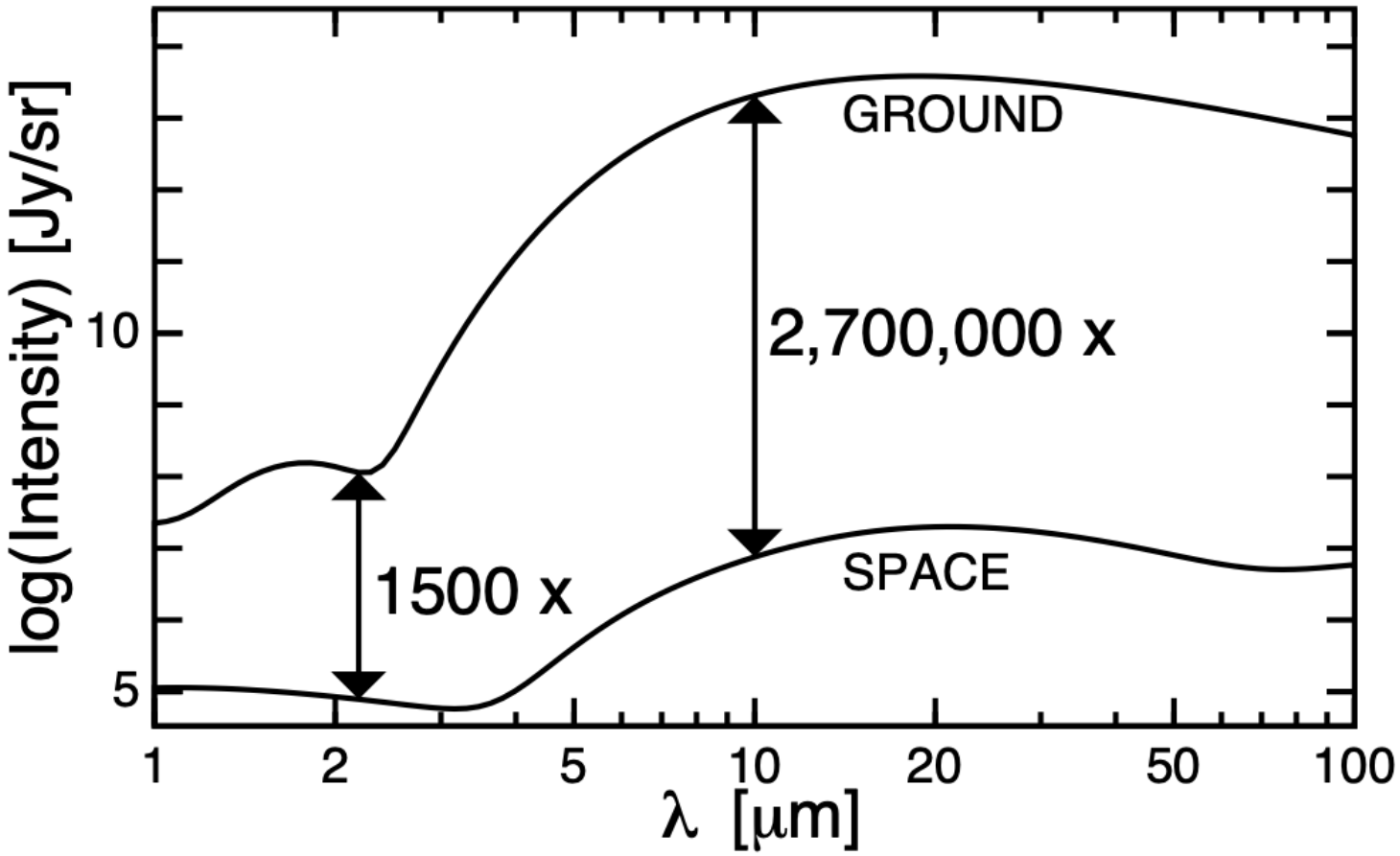
- **ENTIRE** Instruments must be cooled within a cryostat to operate.
- This is why adaptive secondaries help with thermal noise.



Infrared Astronomy Challenges

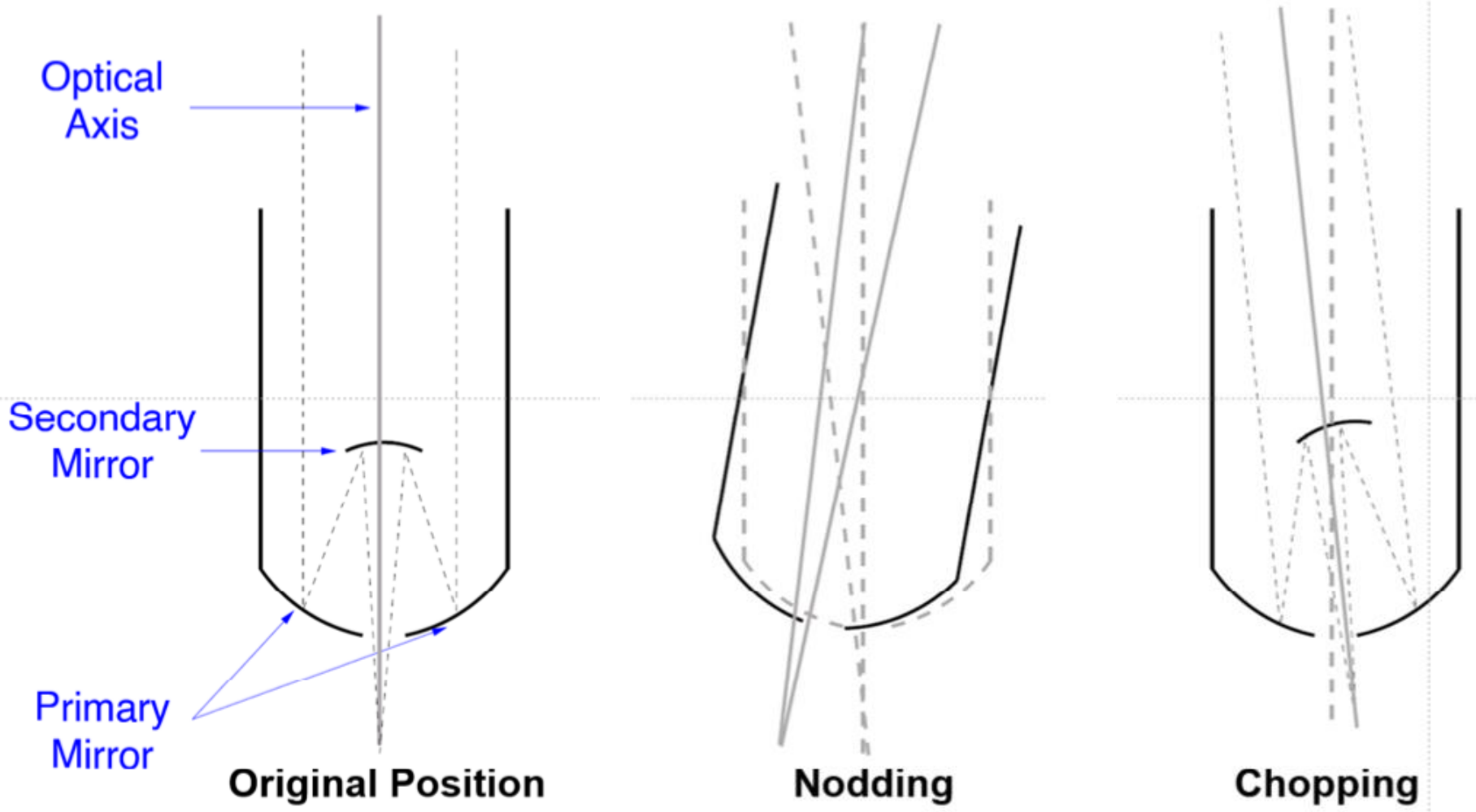
The sky is so much brighter in the infrared than the optical. Even in space there is ambient infrared background.

At most infrared wavelengths it is always your dominant noise source and it changes over time.



The Infrared Sky (Review Article Wright+ 2005)

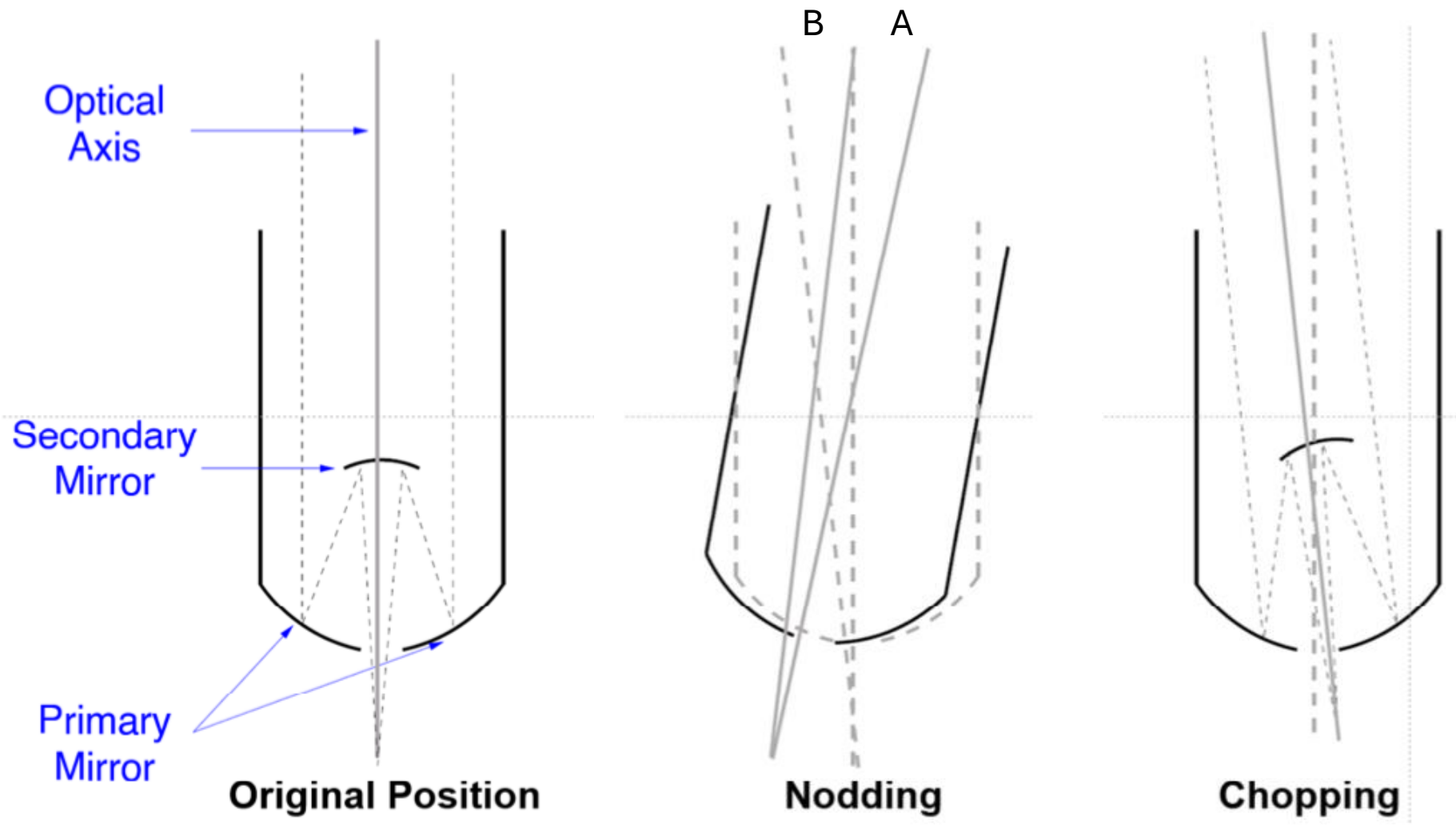
Nodding and Chopping to Remove Noise



Unlike the optical, you need to physically move the telescope to subtract sky and sometimes background from your instrument.

Moens+ 2026

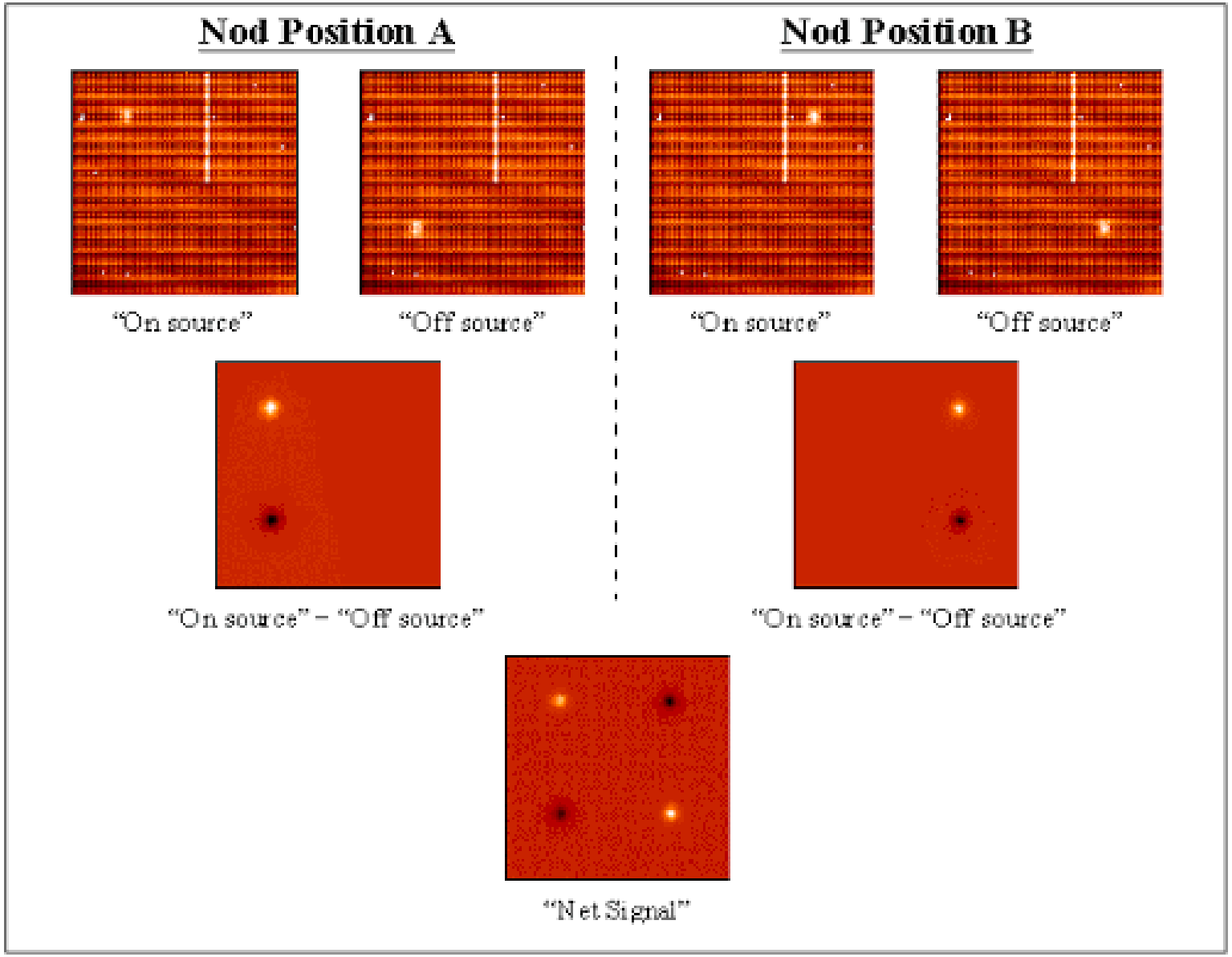
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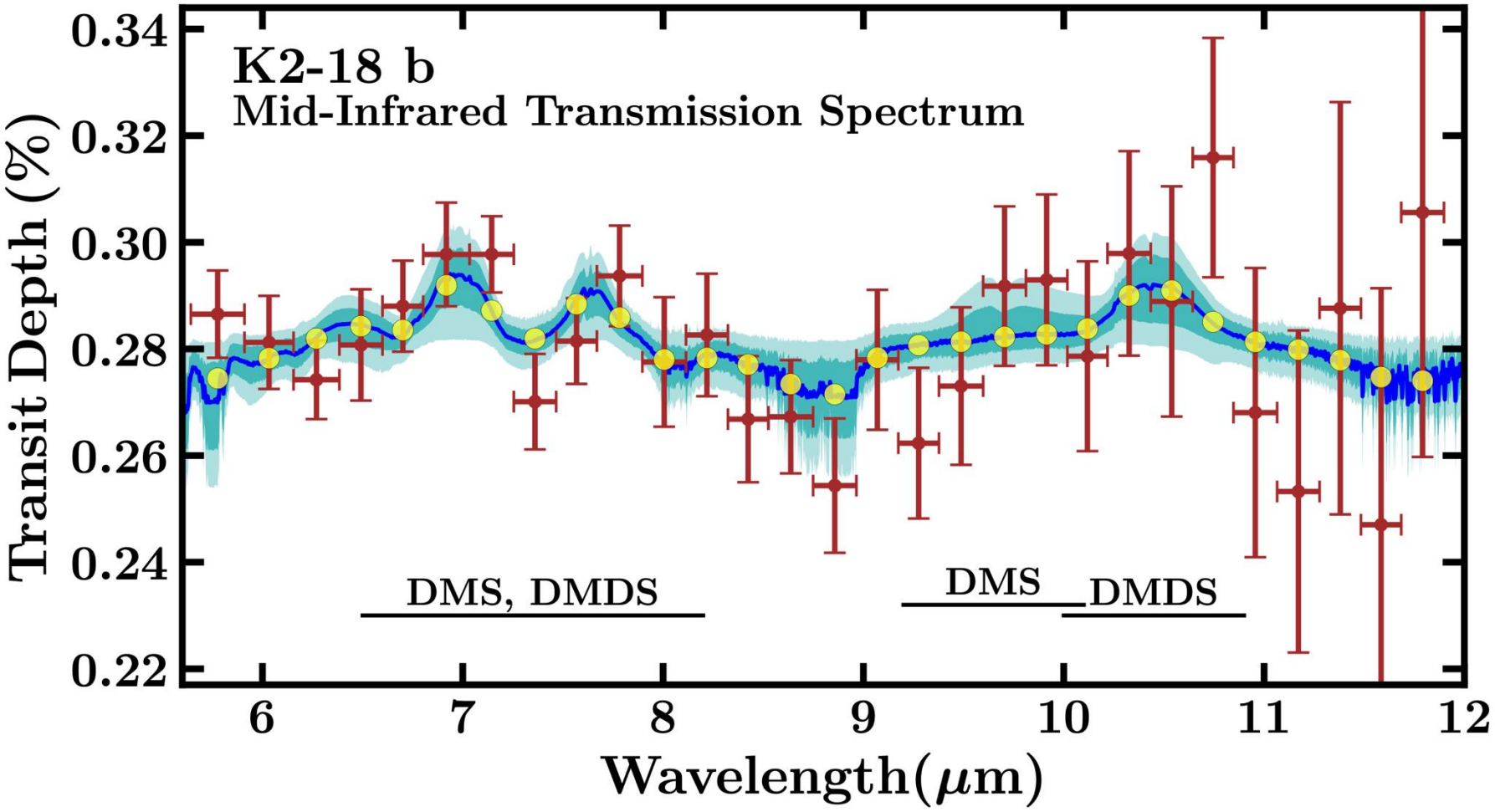
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Moens+ 2026

Nodding and Chopping to Remove Noise



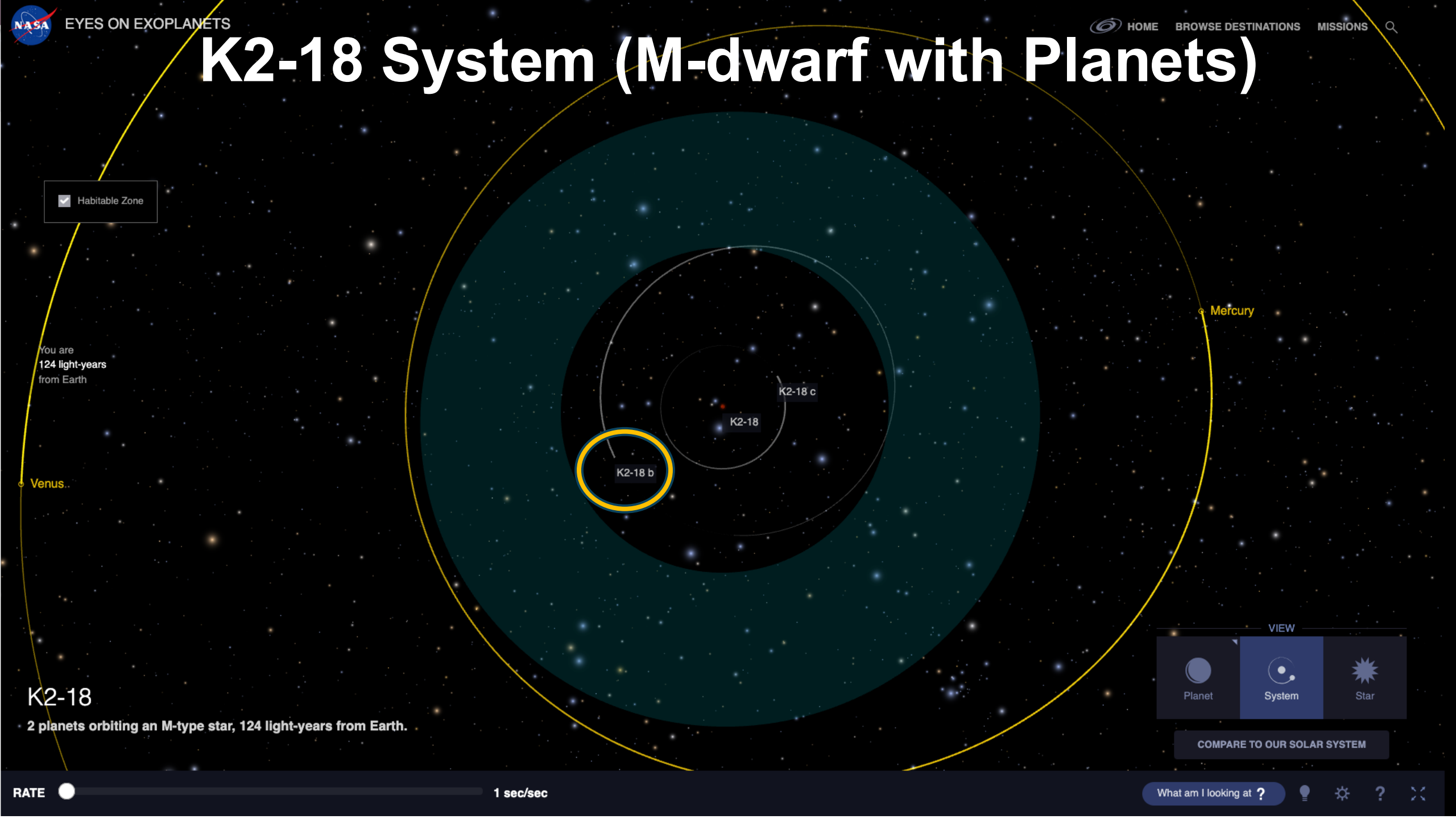
ASTR 302: Fitting Astronomical Data



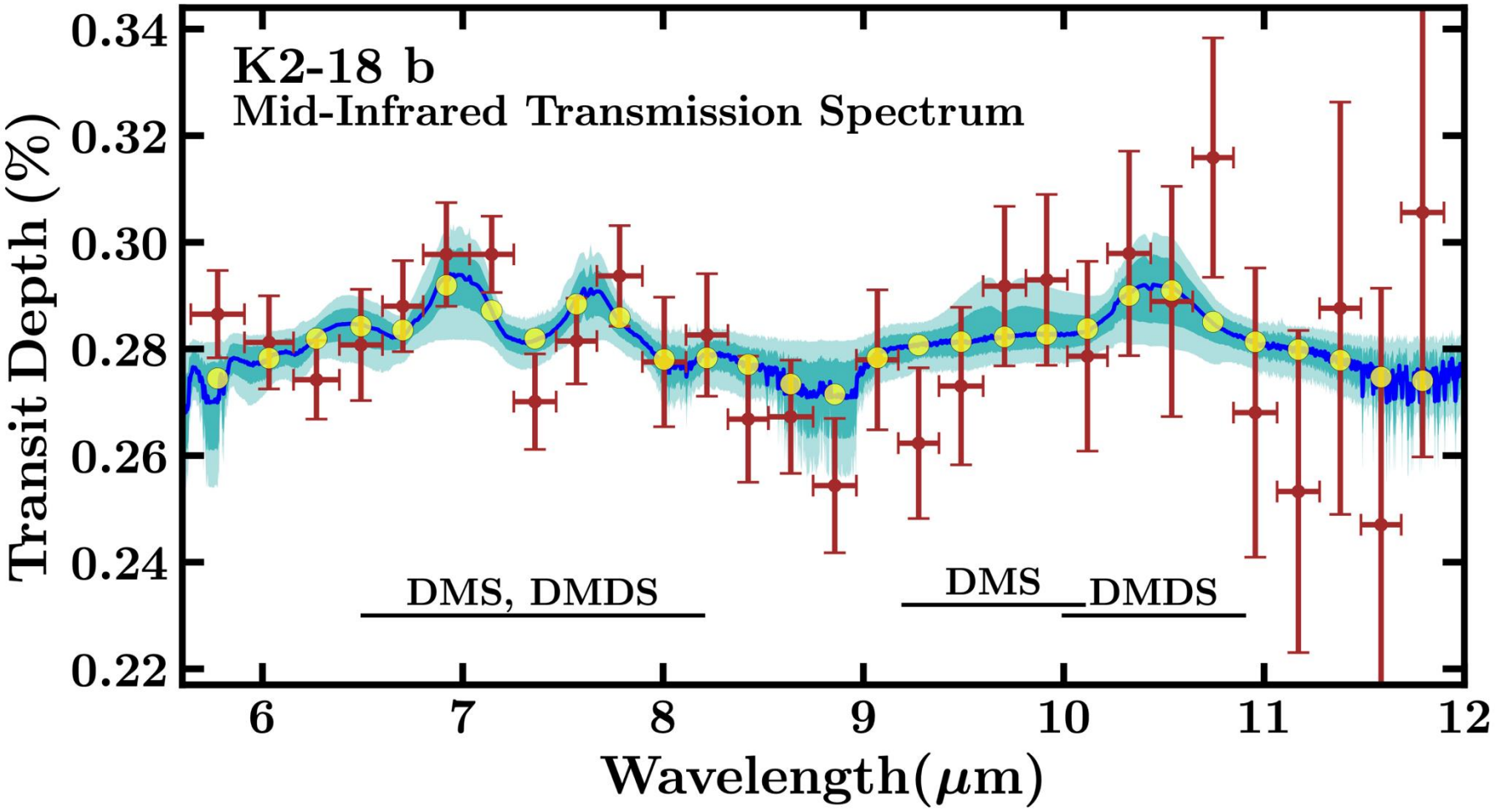
DMS - Dimethyl Sulfide
DMDS – Dimethyl Disulfide

Madhusudhan+ 2025

https://www.npr.org/2025/04/25/g-s1-62610/biosignatures-k2-18b-james-webb-exoplanet-doubt


NASA
EYES ON EXOPLANETS
HOME
BROWSE DESTINATIONS
MISSIONS
K2-18 System (M-dwarf with Planets)
Habitable Zone
You are
124 light-years
from Earth
Venus
Mercury
K2-18 c
K2-18
K2-18 b
VIEW
Planet
System
Star
COMPARE TO OUR SOLAR SYSTEM
K2-18
2 planets orbiting an M-type star, 124 light-years from Earth.
RATE
1 sec/sec
What am I looking at ?

Claim of Biosignature Evidence in K2-18



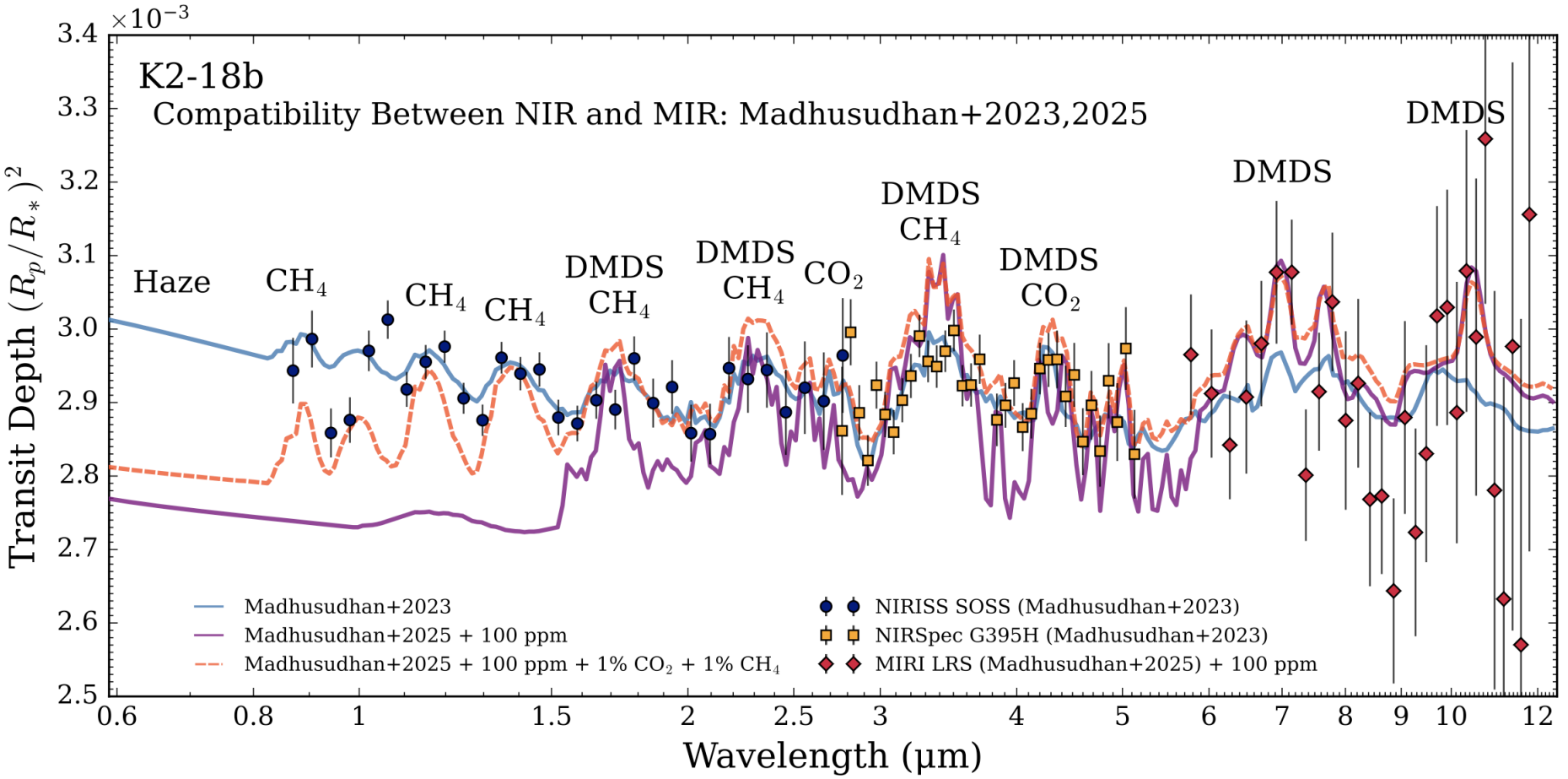
DMS - Dimethyl Sulfide
DMDS – Dimethyl Disulfide

Madhusudhan+ 2025

This biosignature claim is dependent on this planet being a true ocean world.

No water no algae

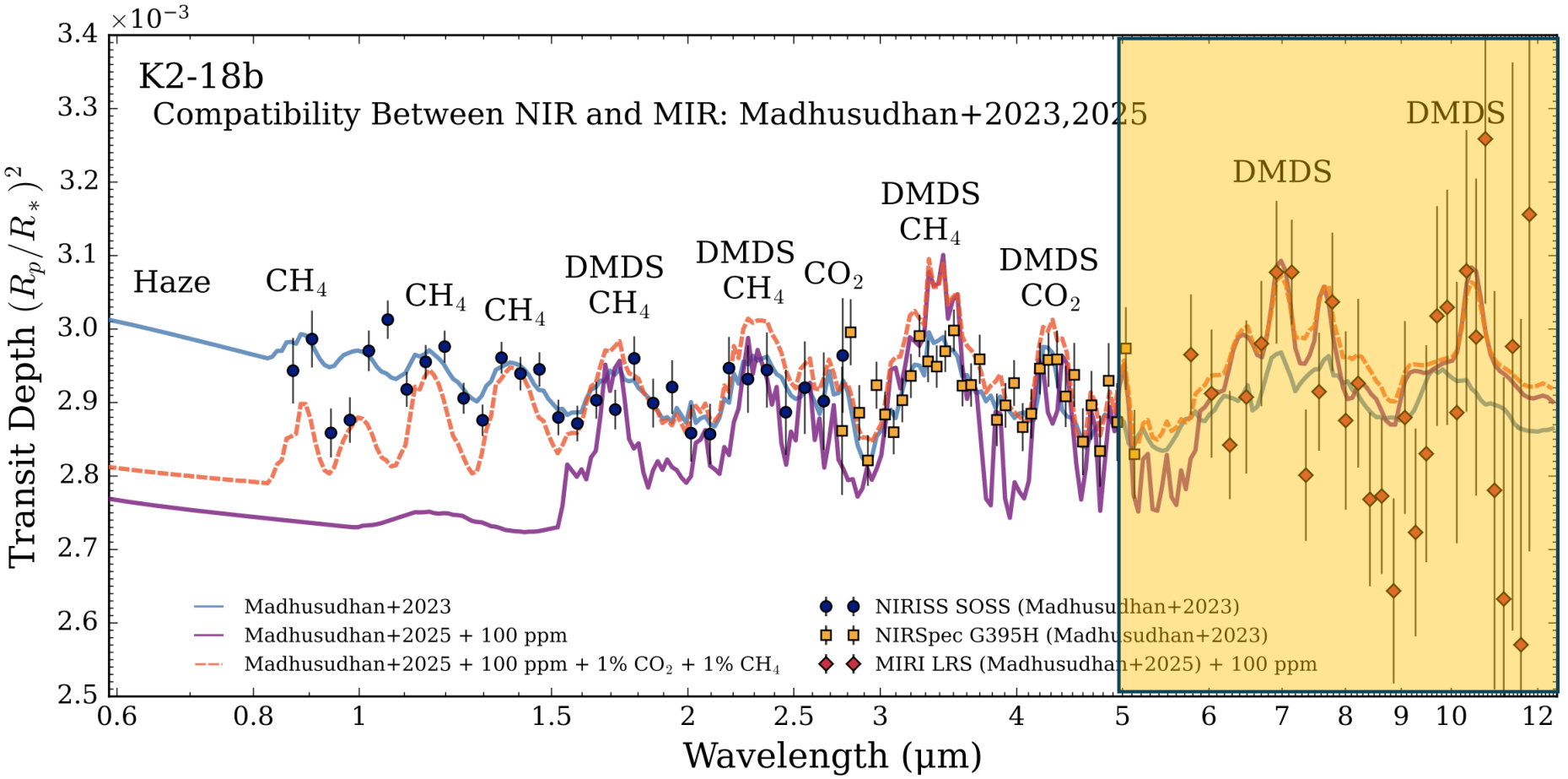
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Schmidt+ 2025

When broader wavelength information is included,
the significance of the DMDS/DMS decreases

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